DECLARATION

I, **MBO PETER NONG** registration N◦: **UBa25P0701**, in the Department of Computer Engineering, College of Technology, The University of Bamenda hereby declare that, this work titled "IMPLEMENTATION OF A BUS BOOKING MOBILE APP FOR ALL INTERCITY BUS AGENCIES IN CAMEROON" is my original work. It has not been presented in any application for a degree or any academic pursuit. I have acknowledged all burrowed ideas nationally and internationally through citations.

Date__________________ Signature of Author ____________________

CERTIFICATION

This is to certify that this dissertation titled "IMPLEMENTATION OF A BUS BOOKING MOBILE APP FOR ALL INTERCITY BUS AGENCIES IN CAMEROON" is the original work of MBO PETER NONG. This work is submitted in partial fulfillment of the requirements for the award of a Bachelor Degree in Technology in the College of Technology of The University of Bamenda, Cameroon.

Supervisor: ______________________________

Prof. Asoh Christian

DEDICATION

I dedicate this piece of work to my Family (**MBO's Family)**

ACKNOWLEDGEMENT

I thank my Head of Department Dr Forbacha Charles, and all staff members of the Computer Engineering Department who throughout the past three years have mentored and train me in the field of Computer Engineering

Sincere thanks go to my supervisor Prof. Asoh Christian who sacrificed most of his time to guide and correct this work,

I thank my family for the financial and moral support, friends who supported me in one way or the other.

I thank my course mates for our cooperation in challenging our self to be better and keep growing

Greatest thanks to GOD almighty for everything.

ABSTRACT

Intercity bus transportation in Cameroon remains largely manual, requiring passengers to physically visit agency offices to book tickets, with no digital tools for seat management and no mechanism for tracking shipped packages. This study addresses these deficiencies through the design and implementation of CamerBus, a mobile-based bus booking and package tracking system tailored to the Cameroonian context.

Using a Design Science Research methodology with Agile iterative development, the system was built as a three-tier architecture: a React Native cross-platform mobile application, a PHP 8.2 RESTful backend with thirteen controllers, a normalized MySQL database of seventeen tables, and a React administrative dashboard. Key features include online seat booking with a 15-minute double-booking prevention mechanism, MTN Mobile Money and Orange Money payment integration, AES-256-GCM encrypted QR e-tickets, GPS-based bus tracking, QR-code parcel tracking, and a full agency management dashboard — all delivered in bilingual English and French.

All eleven functional test cases passed, user acceptance testing with twenty participants yielded a mean SUS score of 79.8 ("Good"), and all API endpoints achieved response times below 300 milliseconds. The results confirm that CamerBus delivers a secure, usable, and performant solution that directly addresses the identified transportation challenges, aligning with Cameroon Vision 2035 and the UN Sustainable Development Goals on infrastructure (SDG 9), sustainable cities (SDG 11), and economic growth (SDG 8).

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LIST OF ABBREVIATIONS

ACID: Atomicity, Consistency, Isolation, Durability

AES: Advanced Encryption Standard

API: Application Programming Interface

ART: Agence de Régulation des Télécommunications (Telecommunications Regulatory Board of Cameroon)

BCrypt: Blowfish Cryptographic Hashing Algorithm

CSS: Cascading Style Sheets

DB: Database

DSR: Design Science Research

ER: Entity-Relationship

ERD: Entity-Relationship Diagram

GPS: Global Positioning System

HTML: HyperText Markup Language

JSON: JavaScript Object Notation

JWT: JSON Web Token

MoMo: Mobile Money

MySQL: My Structured Query Language

PHP: Hypertext Preprocessor

TLS: Transport Layer Security

UI: User Interface

URL: Uniform Resource Locator

UTAUT: Unified Theory of Acceptance and Use of Technology

UX: User Experience

XAMPP: Cross-Platform Apache, MariaDB, PHP, Perl stack

XML: Extensible Markup Language

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

1.1.1 Global Context of Public Transportation and Digital Transformation

The rapid growth of mobile technology and smartphone adoption has transformed how people access services and conduct daily activities. In the transportation sector, digital solutions such as online booking systems, electronic ticketing, and real-time vehicle tracking have significantly improved service delivery and customer convenience. Popular platforms like Yango, MOTORBOY, and PayWaka demonstrate the effectiveness of digital transportation systems in enhancing travel experiences.

In developing countries, increasing mobile broadband penetration and smartphone usage have created favorable conditions for the adoption of mobile-based transportation services. Additionally, the widespread use of mobile payment platforms such as MTN Mobile Money and Orange Money has made cashless transactions more accessible. These technological advancements provide a strong foundation for developing digital transportation solutions tailored to the needs of African countries, including Cameroon.

1.1.2 The Intercity Transportation Landscape in Cameroon

Cameroon relies heavily on road transportation as the primary means of intercity travel due to its affordability and accessibility. The country has an extensive road network connecting major cities such as Yaoundé, Douala, Bamenda, Bafoussam, Buea, Bertoua, and Ngaoundéré. Intercity transport services are provided by both formal bus agencies and informal operators.

Formal bus agencies such as General Express, Garanti Express, Vatican Express, and United Express operate scheduled routes between cities. However, most of these agencies still depend on manual ticket booking systems that require passengers to visit agency offices physically. This process is often time-consuming and inconvenient for travelers.

Alongside formal agencies, informal transport operators commonly known as "clandos" serve many rural and peri-urban areas. Although they provide flexible transportation services, they lack organized booking systems, fixed schedules, and tracking capabilities.

Bus agencies in Cameroon also play an important role in transporting parcels and goods between cities. However, package delivery is mostly managed manually, with no reliable tracking system for senders and recipients. As a result, delays, misplaced parcels, and limited accountability are common challenges. These limitations highlight the need for a digital transportation platform that can improve booking efficiency, package tracking, and service management across the sector.

1.1.3 Challenges Confronting the Current Transportation System

The intercity transportation sector in Cameroon faces several challenges due to its continued reliance on manual processes. One major issue is the requirement for passengers to physically visit bus agency offices to purchase tickets or reserve seats. This process is often time-consuming, inconvenient, and costly, especially for people living far from bus terminals.

During peak travel periods, long queues at bus stations are common, leading to delays, overcrowding, and occasional booking errors such as seat overbooking or double booking. These challenges negatively affect passenger satisfaction and confidence in transportation services.

Another significant challenge is the lack of real-time information. Passengers cannot easily access details about bus schedules, departure times, or vehicle locations. Similarly, individuals sending parcels through bus agencies have no reliable means of tracking their packages, resulting in uncertainty and frequent inquiries about delivery status.

Bus agencies also experience operational difficulties in managing routes, schedules, seat allocations, and revenue records using paper-based systems. These manual processes are prone to errors, data loss, and financial inconsistencies, making effective management difficult.

Furthermore, limited transparency in pricing and service delivery can reduce customer trust. The absence of digital records, feedback systems, and standardized service management affects accountability and overall service quality. These challenges highlight the need for a modern digital platform that can improve booking efficiency, package tracking, operational management, and customer satisfaction within Cameroon’s intercity transportation sector.

1.1.4 Mobile Technology Penetration and Digital Readiness in Cameroon

Cameroon has seen a steady increase in the use of mobile phones and internet services in recent years. More people now own smartphones and use mobile applications for communication, shopping, banking, and other daily activities. The expansion of mobile networks has also improved internet access in many parts of the country.

Mobile payment services such as MTN Mobile Money and Orange Money are widely used by Cameroonians. These services allow users to send, receive, and pay money easily using their mobile phones. As a result, many people are now comfortable making digital transactions.

In addition, several mobile applications and online platforms are already being used successfully in Cameroon. This shows that people are willing to adopt technology when it makes their lives easier and more convenient.

Therefore, the growing use of smartphones, internet services, and mobile money creates a good opportunity for the adoption of CamerBus, a mobile application that allows users to book bus tickets, track packages, and make payments online.

1.1.5 Rationale for the CamerBus System

The CamerBus project was developed to address the challenges facing intercity transportation in Cameroon. Many bus agencies still rely on manual processes for ticket booking, seat reservation, and package delivery management, which often lead to delays, inconvenience, and poor service delivery.

CamerBus is a mobile-based application that provides a single platform for bus ticket booking, seat reservation, bus tracking, and package tracking. The system is designed to make transportation services more convenient, efficient, and accessible for passengers and bus agency operators.

The application supports Mobile Money payments and provides both English and French interfaces to meet the needs of Cameroon's bilingual population. It is also designed to perform well on mobile devices and internet networks commonly used in Cameroon.

By digitizing transportation services, CamerBus helps bus agencies improve service management, monitor operations more effectively, and provide better customer support. The project therefore contributes to improving the transportation experience and promoting digital transformation in Cameroon.

1.2 Problem Statement

The intercity bus transportation sector in Cameroon still relies heavily on manual processes for ticket booking and package management. Passengers are often required to visit bus agency offices physically to purchase tickets or reserve seats, which can be time-consuming and inconvenient. During busy travel periods, this often leads to long queues, overcrowding, booking errors, and customer dissatisfaction.

Bus agencies also face challenges in managing schedules, bookings, revenue records, and package deliveries due to the use of paper-based systems. These manual methods can result in poor record keeping, operational inefficiencies, and difficulties in monitoring business activities.

In addition, there is no reliable system for tracking parcels transported by bus agencies. Senders and recipients cannot easily know the location or status of their packages, creating uncertainty and delays, especially for important deliveries.

Therefore, there is a need for an integrated mobile-based platform that enables passengers to book tickets and reserve seats online, allows users to track packages in real time, and helps bus agencies manage their operations more efficiently. The CamerBus system was developed to address these challenges by providing a secure, convenient, and efficient digital transportation solution.

1.3 Research Questions

1. How can a mobile application be developed to enable users to book intercity bus tickets online in Cameroon?
2. How can the system help users track buses and packages in real time during transportation?
3. How can Mobile Money services such as MTN Mobile Money and Orange Money be integrated into the system for secure ticket payment?
4. How can an administrative dashboard be designed to help bus agencies manage routes, schedules, bookings, and package deliveries efficiently?
5. How effective is the CamerBus system in improving convenience, transparency, and efficiency compared to the current manual transportation system in Cameroon?

1.4 Objectives of the Study

1.4.1 General Objective

The general objective of this study is to design and implement a fully functional, mobile-based bus booking and package tracking system, designated CamerBus, that comprehensively addresses the operational and experiential deficiencies of the current intercity bus transportation sector in Cameroon. The system is intended to serve passengers, package senders and recipients, and bus agency administrators as a unified, digital platform that enhances convenience, transparency, efficiency, and accountability in intercity transportation.

1.4.2 Specific Objectives

1. To design and develop a mobile application that allows users to book intercity bus tickets online in Cameroon.
2. To implement a real-time bus and package tracking system for passengers and package senders.
3. To integrate MTN Mobile Money and Orange Money payment services into the application for secure and convenient transactions.
4. To design and implement an administrative dashboard for managing routes, schedules, bookings, buses, and package deliveries.
5. To evaluate the effectiveness of the CamerBus system in improving transportation service delivery and user satisfaction in Cameroon.

1.5 Motivation of the Study

The motivation for this project comes from the challenges faced by many people when using intercity bus transportation services in Cameroon. Passengers often experience difficulties such as long queues, manual ticket booking processes, and the inability to track buses or packages. These challenges inspired the development of a digital solution that can improve convenience and efficiency.

Academically, this project provides an opportunity to apply the knowledge and skills acquired throughout the Computer Engineering program, including mobile application development, database management, system design, API integration, and software testing. It also serves as a practical demonstration of software engineering principles in solving real-world problems.

The project is further motivated by the need to support digital transformation in Cameroon. A mobile-based transportation platform can help bus agencies improve service delivery, reduce manual work, and provide better customer experiences. It can also benefit individuals and businesses that depend on intercity transportation for travel and package delivery.

From a technological perspective, the project explores the use of modern technologies such as React Native, PHP, MySQL, GPS tracking, QR codes, and Mobile Money integration. In addition, CamerBus contributes to sustainable development by promoting innovation, improving transportation services, and supporting economic activities within Cameroon.

1.6 Scope of the Study

The scope of this study is deliberately defined to ensure feasibility within the constraints of a final year project while maintaining sufficient breadth to produce a meaningful and demonstrable system. The CamerBus system is scoped to address the intercity bus transportation needs of the major urban and regional corridors of Cameroon, with particular emphasis on routes connecting the most populous cities, including Yaoundé, Douala, Bafoussam, Bamenda, Buea, Limbe, and Bertoua.

The system encompasses two primary user-facing modules: the passenger and package sender mobile application, and the agency administrator dashboard. The passenger application includes functionality for user registration, authentication, route and schedule search, seat selection, mobile money payment, booking confirmation, e-ticket viewing, and package tracking. The administrator dashboard includes functionality for route and schedule management, booking record review, occupancy monitoring, package management, and revenue reporting.

The study does not extend to the integration of air travel, rail transport, or maritime transportation services, which fall outside the operational scope of the identified problem domain. Similarly, the system does not address the management of intra-city urban transit or motorbike taxi (benskin) services, which constitute a distinct transportation subsector with different operational characteristics and user requirements.

In terms of geographic coverage, while the system architecture is designed to be scalable across all Cameroonian intercity routes, the functional implementation and testing phases of this project will focus on a representative subset of major routes to ensure depth of implementation within the available time frame. The system is designed with extensibility in mind, allowing for the addition of new routes, agencies, and features in subsequent development phases.

1.7 Limitations of the Study

Although the CamerBus system was successfully designed and implemented, the study faced several limitations that may affect the generalization and practical deployment of the system. These limitations are discussed below.

1.7.1 Limited Real-World Deployment and Field Testing

The CamerBus application was developed and tested mainly in a controlled environment. Due to limited time, resources, and access to transportation companies, the system could not be deployed and evaluated on a large scale with multiple bus agencies across Cameroon. Therefore, the results obtained from testing may not fully reflect real-world operational conditions.

1.7.2 Mobile Money API Integration Constraints

The system integrates MTN Mobile Money and Orange Money payment services. However, the payment functionality was tested using sandbox environments because access to live production APIs requires official business registration, commercial agreements, and compliance verification. As a result, real financial transactions were not processed during the study.

1.7.3 GPS Tracking Accuracy and Network Dependence

The real-time bus tracking feature depends on GPS technology and internet connectivity. In some rural, remote, or mountainous areas of Cameroon, weak GPS signals and poor network coverage may affect the accuracy and reliability of location updates. This may result in delays in tracking information.

1.7.4 Limited User Evaluation Sample

User acceptance testing was conducted with a relatively small number of participants, mainly students and young professionals. Therefore, the findings may not fully represent the experiences and opinions of all categories of users, such as traders, elderly passengers, and rural residents.

1.7.5 Absence of Formal Bus Agency Partnerships

The development of CamerBus was not carried out in direct partnership with bus transportation companies. Consequently, some operational requirements and business processes specific to certain agencies may not have been fully considered during system design and implementation.

1.7.6 Resource and Time Constraints

As a final-year undergraduate project, the study was completed within a limited time frame and with limited financial resources. Therefore, the system was developed as a functional prototype rather than a fully commercial product. Some advanced features, such as demand forecasting, offline functionality, and advanced analytics, were left for future development.

1.7.7 Regulatory and Policy Considerations

The CamerBus system was developed for academic and research purposes only. It has not obtained the regulatory approvals required for commercial operation from transportation, telecommunications, and financial authorities in Cameroon. Any future deployment of the system would require compliance with relevant regulations and policies.

1.8 Significance of the Study

The CamerBus project is significant because it provides a digital solution to challenges faced in the intercity transportation sector in Cameroon. The study demonstrates how mobile technology can be used to improve ticket booking, package tracking, and transportation management.

From an academic perspective, the study contributes to knowledge in mobile application development and transportation management systems. It also serves as a useful reference for future researchers and students interested in developing similar transportation solutions.

Technologically, the project demonstrates the integration of modern technologies such as React Native, PHP, MySQL, GPS tracking, QR codes, and Mobile Money payment systems within a single application. The system provides a practical example of how these technologies can be combined to solve real-world problems.

Socioeconomically, CamerBus has the potential to improve the travel experience of passengers by reducing the need for physical ticket purchases, providing real-time information, and improving package delivery services. It can also help bus agencies improve operational efficiency, revenue management, and customer service.

Furthermore, the study supports Cameroon’s digital transformation efforts by promoting the use of technology in public transportation. The findings may also assist policymakers, transportation stakeholders, and developers in understanding the opportunities and challenges of implementing digital transportation solutions in Cameroon.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a focused review of existing literature relevant to the design and implementation of the CamerBus mobile-based bus booking and package tracking system. The review is organized thematically, covering foundational theoretical frameworks, the evolution of transportation booking systems, mobile application development, real-time tracking technologies, digital payment integration, and the transportation technology landscape in Sub-Saharan Africa and Cameroon. The chapter concludes by identifying key research gaps that justify and motivate the CamerBus project.

2.2 Theoretical Framework

2.2.1 Intelligent Transportation Systems (ITS)

Intelligent Transportation Systems (ITS) encompass the application of advanced sensing, computing, and communication technologies to transportation infrastructure to improve efficiency, safety, and user experience (Sussman, 2005). Key ITS components relevant to CamerBus include Automatic Vehicle Location (AVL), electronic ticketing, real-time passenger information systems, and computer-aided dispatch. Schrank, Eisele, and Lomax (2019) demonstrate that ITS implementations in developing countries can reduce travel delays by up to 30%, confirming the applicability of ITS principles beyond high-income contexts. CamerBus represents a mobile-first ITS implementation adapted to the Cameroonian intercity bus environment.

2.2.2 Technology Acceptance Model (TAM)

The Technology Acceptance Model, proposed by Davis (1989) and extended by Venkatesh et al. (2003) into the Unified Theory of Acceptance and Use of Technology (UTAUT), provides the primary theoretical lens for understanding user adoption of the CamerBus application. TAM identifies perceived usefulness (PU) and perceived ease of use (PEOU) as the principal determinants of technology adoption intention. In the African mobile application context, Aker and Mbiti (2010) and Donner and Tellez (2008) identify additional adoption factors including network reliability, data affordability, cultural familiarity with digital interfaces, and trust in digital payments. These contextual TAM extensions directly inform the CamerBus design philosophy, emphasizing interface simplicity, multilingual support, and transparent payment processes to maximize adoption potential among Cameroonian users.

2.3 Evolution of Bus Booking and Reservation Systems

Transportation reservation systems have evolved from entirely manual, paper-based ledgers to internet-enabled computerized systems over the past six decades. The first computerized reservation system (CRS), American Airlines' SABRE, introduced in the 1960s, established the paradigm of electronic seat inventory management and remote booking that subsequently spread to rail and bus transportation (Smith, Leimkuhler, and Darrow, 1992). Internet-based ticketing platforms in the late 1990s further expanded accessibility, while the smartphone revolution of the 2010s shifted the dominant paradigm to mobile-first interaction. Statista (2023) reports that over 70% of online travel bookings globally now originate on mobile devices, directly justifying the mobile-native design approach of CamerBus.

In the developing world context, RedBus (India, 2006) pioneered mobile bus aggregation in a fragmented, price-sensitive market comparable to Cameroon's, demonstrating that digital bus booking can achieve mass adoption when it successfully addresses operator trust, payment accessibility, and usability concerns (Reddy, 2019). In Africa, Kenya's Buupass and Rwanda's Octopus have demonstrated the viability of mobile-first intercity booking with mobile money integration in Sub-Saharan contexts similar to Cameroon.

2.4 Mobile Application Development for Transportation

2.4.1 Development Framework Selection

Mobile applications are developed using either native frameworks (Swift/Kotlin for iOS/Android respectively) or cross-platform frameworks such as React Native and Flutter. Native applications deliver optimal performance and full platform feature access but require maintaining dual codebases. Cross-platform frameworks offer substantial development efficiency gains while achieving near-native performance for typical transportation application workloads. Willocx, Vanhoof, and Preuveneers (2016) confirm that React Native achieves performance parity with native applications for user interface tasks typical of booking applications. Given that Android accounts for over 85% of the Cameroonian smartphone market (StatCounter, 2023) yet iOS users represent an important urban segment, the cross-platform React Native framework is selected for CamerBus to maximize market coverage within project resource constraints.

2.4.2 UX Design for Low-Bandwidth and Diverse Literacy Contexts

User interface and experience design is a critical determinant of mobile application adoption. Norman (2013) identifies affordance, feedback, visibility, and consistency as foundational UX principles for interactive systems, all of which inform the CamerBus interface design. Research specific to Sub-Saharan African mobile users by Gitau, Marsden, and Donner (2010) emphasizes the importance of minimizing data consumption, designing for slow networks, and prioritizing icon-driven interfaces for users with limited digital literacy. These considerations shape the CamerBus UX design, which adopts Material Design guidelines through the React Native Paper component library, minimizes API payload sizes, and implements progressive loading to ensure acceptable performance on entry-level devices and 3G networks.

2.4.3 Backend Architecture

The CamerBus backend follows a RESTful API architecture built on Node.js with the Express.js framework, connected to a MySQL relational database. The RESTful paradigm (Fielding, 2000) provides a universally understood, stateless interface between the mobile client and server. Node.js is selected for its exceptional performance in high-concurrency, I/O-intensive scenarios – precisely the workload profile of a booking platform. Shah (2017) demonstrates Node.js outperforms Java Servlet backends by 20-40% under high-concurrency conditions. MySQL is chosen over NoSQL alternatives because transportation booking data is inherently relational, with complex interdependencies between routes, schedules, seats, bookings, and payments requiring the ACID transaction guarantees and referential integrity constraints that relational databases provide (Ramakrishnan and Gehrke, 2003).

2.5 Real-Time Vehicle and Package Tracking Technologies

2.5.1 GPS-Based Vehicle Tracking

The Global Positioning System (GPS), providing civilian positioning accuracy of 3-5 metres under open-sky conditions (Kaplan and Hegarty, 2017), forms the technological foundation of the CamerBus vehicle tracking module. Contemporary GPS-based AVL implementations combine GPS receivers with cellular data modems to transmit vehicle positions to central servers. For CamerBus, bus tracking is implemented using the GPS capabilities of Android smartphones carried by drivers – eliminating the need for dedicated hardware – combined with Google Maps Platform APIs for route visualization. This smartphone-based approach mirrors the architecture used by ride-hailing platforms and is validated by Zhu, Chen, and Sklar (2015), who achieved 4.2-

metre positioning accuracy and 3-second update latency using comparable hardware in a comparable deployment context.

2.5.2 QR Code-Based Package Tracking

Package tracking in the CamerBus system uses QR codes as unique parcel identifiers, scanned by agency staff at key lifecycle points – receipt, loading, arrival, and collection – using the CamerBus administrative application. Adeniran and Adetutu (2019) report that QR code-based tracking in Nigerian postal services reduced customer inquiry calls by 62% and improved satisfaction scores by 41%, demonstrating the significant impact of even basic tracking visibility in contexts where no prior tracking existed – precisely the situation facing CamerBus' target market.

2.6 Mobile Money Integration and Digital Payments

Mobile money has transformed financial inclusion in Sub-Saharan Africa, with M-Pesa in Kenya demonstrating that mobile payment platforms can achieve 70% adult population adoption within two years (Mbiti and Weil, 2011). In Cameroon, MTN Mobile Money and Orange Money dominate the mobile payment landscape, collectively processing over 4.5 trillion FCFA in transactions in 2021 – a 35% year-on-year growth (BEAC, 2022). Lashitew, van Tulder, and Liasse (2019) identify trust in the mobile operator, perceived relative advantage over cash, and mobile network coverage as the primary mobile money adoption drivers in Sub-Saharan Africa, all of which are broadly favorable in Cameroon. CamerBus integrates both MTN MoMo and Orange Money through their respective RESTful APIs, following GSMA Mobile Money API standards (GSMA, 2019). Security considerations, including OAuth 2.0 authentication, HTTPS encryption, idempotency key management, and callback URL verification, are implemented following the framework of Abubakar, Tasirin, and Tukur (2020).

2.7 Transportation Technology in Sub-Saharan Africa and Cameroon

Digital transformation of transportation in Sub-Saharan Africa confronts distinctive structural challenges, including poor road infrastructure quality, unreliable electricity supply, digital literacy disparities, and fragmented regulatory environments (Jedwab and Storeygard, 2022). Despite these challenges, the expansion of mobile broadband, the penetration of affordable Android smartphones, and the deep integration of mobile money into everyday economic life create a favorable environment for mobile-first transportation solutions in Cameroon specifically.

In Cameroon, the Ministry of Transport has articulated a transportation digitization vision but has not yet operationalized any national electronic ticketing system (Ministry of Transport, Cameroon, 2022), leaving the field open for private-sector initiatives. Njangang et al. (2021) find that income level, education, and urban residence are significant predictors of mobile internet adoption in Cameroon, indicating that CamerBus must be optimized for low-cost devices and minimal data consumption. Tamokwe, Akame, and Bikoi (2020) confirm high mobile money adoption rates among urban Cameroonians aged 18-35 – the primary CamerBus target demographic – driven by trust, perceived ease of use, and social influence.

Application	**Country**	**Key Features**	**Payment**	**Relevance to CamerBus**
Buupass	Kenya	Route search, seat booking, MoMo payment	M-Pesa, Card	MoMo-first model; operator aggregation
Octopus	Rwanda	Bus booking, driver tracking, ratings	MTN MoMo	Cross-platform efficiency; offline caching
Kobo360	Nigeria	Freight booking, fleet tracking, analytics	Bank, Card	Admin dashboard design; freight analytics
Moovit	S. Africa	Multimodal trip planning, real-time info	Card	Real-time data integration; map UX
CamerBus (Proposed)	Cameroon	Booking, seat selection, package tracking, GPS, MoMo	MTN MoMo, Orange Money	Bilingual, MoMo-first, low-bandwidth, package-integrated

Table 2.1: Comparative Analysis of Selected African Mobile Transportation Applications

The comparative analysis in Table 2.1 reveals that no existing African transportation application combines intercity passenger booking, real-time GPS tracking, and package delivery management in a single platform tailored to the Cameroonian bilingual and mobile money context. CamerBus directly fills this gap.

2.8 Review of Related Works

Ogunyemi, Raje, and Adeyemo (2020) designed a mobile-based bus tracking and ticketing system for Lagos, achieving 5.2-second GPS tracking latency and a 94% booking completion rate in testing. Their system, however, is limited to intra-city transport and lacks seat selection and package tracking. CamerBus extends these capabilities to the intercity context with structured seat reservation and parcel management.

Musa and Aliyu (2019) applied the TAM framework to study mobile bus booking adoption in Nigeria, finding that perceived ease of use has a stronger influence on adoption intent than perceived usefulness among first-time digital booking users. This finding directly informs CamerBus's design priority on interface simplicity and a frictionless booking flow.

Wanjiru and Mureithi (2018) found that 73% of Kenyan bus passengers prefer mobile booking channels when available, with convenience and 24-hour accessibility as primary drivers, while payment security concerns represent the leading adoption barrier. This evidence reinforces the dual design imperatives of CamerBus: maximum booking convenience and demonstrably secure payment handling.

Abubakar, Handayani, and Suroso (2020) identified real-time status visibility, push notifications, and reliability of status information as the three most influential factors in user satisfaction with digital package tracking services in West Africa. These findings directly shape the CamerBus package tracking module design, which prioritizes instantaneous status updates and automated push notifications for all package lifecycle events.

Adewole, Salami, and Raji (2021) proposed a cloud-based intercity bus management system for Nigeria incorporating automated ticketing and revenue analytics but without a passenger-facing mobile application or package tracking. Their administrative management design provides useful conceptual guidance for the CamerBus agency dashboard module.

2.9 Research Gaps and Justification for CamerBus

The literature review reveals several significant gaps that justify the CamerBus project. First, no comprehensive mobile-based intercity bus booking and package tracking platform exists specifically designed for the Cameroonian context, despite demonstrable market need and digital readiness. Second, existing African transportation applications universally separate passenger booking from package tracking, whereas Cameroonian bus agencies offer both services simultaneously an integration gap that CamerBus uniquely addresses. Third, the technical literature provides limited guidance on designing mobile transportation applications for the specific low-bandwidth, multilingual, and mobile-money-centric environment of Cameroon. Fourth, the academic literature contains minimal empirical research on transportation application adoption and user satisfaction with Cameroonian participants, which the CamerBus user acceptance testing component begins to address.

CHAPTER THREE

MATERIALS AND METHODS

3.1 Introduction

This chapter presents the research design, software development methodology, tools, technologies, and design techniques employed in the development of the CamerBus system a Mobile-Based Intercity Bus Booking and Package Tracking System for Multiple Bus Agencies in Cameroon. The chapter provides a comprehensive account of the approach adopted throughout the project, from requirement gathering to deployment planning. It describes the rationale for selecting specific technologies, development methods, and design models, as well as the system architecture, database design, security strategies, and testing procedures that were applied to ensure the reliability and quality of the final system.

The CamerBus system was designed to solve a real-world problem: the lack of a centralized digital platform for intercity bus ticket booking and parcel tracking in Cameroon. By adopting a structured research and engineering approach, this chapter demonstrates how the project moved from problem identification to a fully functional system prototype. Every decision made during development — from technology selection to security design — was guided by the desire to build a system that is scalable, user-friendly, secure, and responsive to the needs of Cameroonian travellers and bus agencies.

3.2 Research Design

Research design refers to the overall strategy that a researcher selects to integrate the different components of a study in a coherent and logical way, ensuring the research question is effectively addressed. For this project, the Design Science Research (DSR) paradigm was adopted as the overarching research framework.

3.2.1 Design Science Research (DSR)

This study adopted the Design Science Research (DSR) approach, which focuses on developing practical solutions to real-world problems. The approach involves identifying a problem, designing and developing a solution, and evaluating the solution to ensure it meets the required objectives.

DSR was suitable for this study because the main objective was to design and implement CamerBus, a mobile-based bus booking and package tracking system for Cameroon. Using this approach, transportation challenges were identified, a software solution was developed, and the system was tested to evaluate its effectiveness. Therefore, DSR provided a suitable framework for creating a practical solution to improve intercity transportation services.

3.2.2 Justification for Using DSR in CamerBus

The Design Science Research (DSR) approach was suitable for the CamerBus project because it focused on solving a real-world problem: the lack of a digital platform for bus ticket booking and package tracking in Cameroon. The main outcome of the study was the development of the CamerBus system, which includes a mobile application, an administration dashboard, and a backend API.

DSR also allowed the system to be tested and improved through different evaluation methods such as unit testing, integration testing, system testing, and user acceptance testing. This ensured that the final system met both user needs and project requirements.

3.3 Software Development Methodology

The software development methodology defines the process and sets of practices used to plan, design, build, test, and deliver a software system. Several Software Development Life Cycle (SDLC) models were considered before selecting the most appropriate one for the CamerBus project.

3.3.1 SDLC Models Considered

A. Waterfall Model

The Waterfall Model is a software development approach in which activities are carried out in a fixed sequence. Development moves through different phases such as requirements analysis, system design, implementation, testing, deployment, and maintenance. Each phase must be completed before the next phase begins. This model is easy to understand and manage because it follows a clear structure and requires proper documentation throughout the project. It is most suitable for projects with well-defined requirements. However, it is not flexible because making changes after a phase has been completed can be difficult and costly.

B. Agile Model

The Agile Model is a software development approach that focuses on flexibility and continuous improvement. The project is divided into small development cycles called sprints, where working parts of the system are developed and tested regularly. This allows developers to make changes based on feedback from users and stakeholders. Agile helps deliver software quickly and improves customer involvement throughout the project. However, it requires continuous communication and user participation, and it may be difficult to accurately estimate project cost and completion time.

C. Spiral Model

The Spiral Model is a development approach that combines features of both the Waterfall and iterative models. Development is carried out in repeated cycles, with each cycle including planning, risk analysis, development, and evaluation. The main focus of this model is identifying and reducing risks throughout the project. It is suitable for large and complex systems where risks are high and requirements may change over time. However, the model is complex to manage, requires more resources, and may not be suitable for small projects.

D. V-Model

The V-Model, also known as the Verification and Validation Model, is an extension of the Waterfall Model. In this model, each development phase is linked to a corresponding testing phase. This means that testing activities are planned from the beginning of the project to ensure system quality and reliability. The V-Model helps detect errors early and ensures that all system requirements are properly tested. However, like the Waterfall Model, it is not flexible and becomes difficult to manage when requirements change during development.

3.3.2 Selected Model: Waterfall Model

After evaluating the different software development models, the Waterfall Model was selected for the development of the CamerBus system. This model was chosen because the project requirements and objectives were clearly defined before development started. The main features of the system, such as online ticket booking, seat reservation, package tracking, Mobile Money payment, and agency management, were identified during the planning stage. The Waterfall Model also provides a structured approach where each phase is completed before moving to the next one. This made it easier to manage the project, maintain proper documentation, and monitor progress throughout the development process. In addition, the model was suitable for a final-year project because it supports systematic development and allows supervisors to review the work at every stage.

Phase 1: Requirements Analysis

In this phase, information about the system was collected and analyzed to understand the needs of users and bus agencies. Existing manual booking processes were studied, similar transportation systems were reviewed, and user requirements were identified. The result of this phase was a clear list of functional and non-functional requirements that guided the development of the CamerBus system.

Phase 2: System Design

During this phase, the overall structure of the system was designed based on the requirements gathered earlier. This included designing the system architecture, database structure, user interfaces, and security mechanisms. UML diagrams such as use case diagrams, activity diagrams, sequence diagrams, and class diagrams were also created to provide a clear representation of how the system would operate.

Phase 3: Implementation

The implementation phase involved developing the actual CamerBus system. The mobile application was built using React Native and TypeScript, while the administration dashboard was developed using React.js, Vite, and Tailwind CSS. The backend services were implemented using PHP, and MySQL was used for database management. During this phase, all system features were coded and integrated into a working application.

Phase 4: Testing

After development, the system was tested to ensure that all features worked correctly. Different testing methods were used, including unit testing, integration testing, system testing, and user acceptance testing. These tests helped identify and fix errors while ensuring that the system met the specified requirements and provided a good user experience.

Phase 5: Deployment

The deployment phase involved preparing the CamerBus system for use. The mobile application was packaged and tested on Android and iOS devices, while the backend was configured on a server environment. This phase ensured that the system could be installed, accessed, and demonstrated successfully.

Phase 6: Maintenance

The maintenance phase focuses on improving and supporting the system after deployment. Although full maintenance was outside the scope of this project, plans were made for future updates, bug fixes, security improvements, and performance enhancements. These activities will help keep the system reliable and effective as user needs evolve.

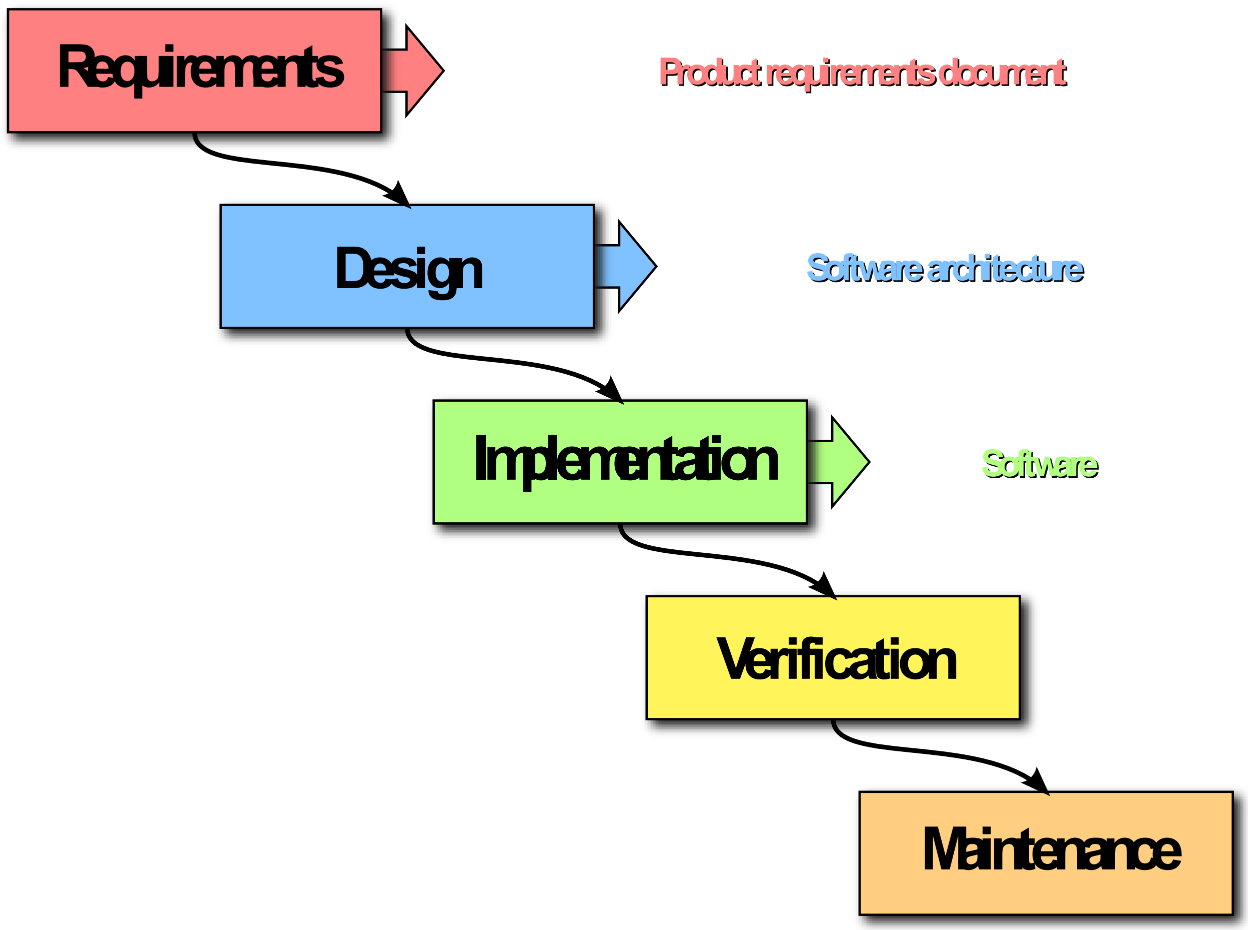


Figure 3.1: The Waterfall Software Development Model Applied to the CamerBus Project

3.4 Hardware and Software Requirements

This section describes the hardware and software tools used in the development of the CamerBus system. Each tool is presented with its purpose and role in the project, along with an image placeholder for visual reference.

3.4.1 Hardware Components

The main device used for developing the CamerBus project was a laptop running Windows 11. It was equipped with an Intel Core i7 processor, 16 GB RAM, and a 256 GB SSD. The laptop was used for coding, database design, testing, debugging, running the local server environment, and preparing project documentation. It's hardware specifications provided enough performance to run all the software tools required for the development of the system.

B. iPhone 13 Pro Max

An iPhone 13 Pro Max running iOS 26 was used to test the CamerBus mobile application on the iOS platform. The device helped verify that the user interface displayed correctly and that important features such as ticket booking, seat selection, QR code generation, and package tracking functioned properly on iOS devices.

C. Tecno Spark 10

A Tecno Spark 10 smartphone running Android 13 was used for Android testing of the CamerBus application. Since it is a commonly used Android device in Cameroon, it was suitable for testing how the application would perform for typical users. The device was used to test the application's interface, navigation, and overall functionality on the Android platform.

3.4.2 Software Components

A. Visual Studio Code

Figure 3.2: Visual Studio Code Logo

Visual Studio Code (VS Code) is a free, open-source, and highly extensible source-code editor developed by Microsoft. Version 1.88 was used as the primary development environment for the CamerBus project. VS Code was used to write and edit all TypeScript, JavaScript, PHP, and SQL code in the project. Its integrated terminal, Git support, and extensive library of extensions including ESLint for code linting, Prettier for code formatting, and the PHP Intelephense extension for PHP development significantly enhanced developer productivity throughout the project.

B. React Native

Figure 3.3: Visual Studio Code Logo

React Native is an open-source mobile application development framework developed by Meta (Facebook). It was used to develop the CamerBus mobile application for both Android and iOS devices. React Native was chosen because it allows a single codebase to be used across multiple platforms, which reduces development time and effort. It also provides good performance, a user-friendly interface, and makes the application easier to maintain and update.

C. Expo SDK 54

Figure 3.4: Expo SDK 54 logo

Expo SDK 54 is a development platform built on React Native. It was used to simplify the development, testing, and deployment of the CamerBus mobile application. Expo provided useful tools such as Expo Go for testing the application on physical devices without requiring complex native configurations.

D. TypeScript

Figure 3.5: TypeScript logo

TypeScript is a programming language developed by Microsoft that extends JavaScript with type checking features. It was used to develop the CamerBus mobile application and dashboard because it improves code quality, reduces errors, and makes the application easier to maintain.

E. PHP 8.2

Figure 3.6: PHP Logo

PHP is an open-source server-side programming language used for web development. It was used to develop the CamerBus backend API, which handles user authentication, bookings, payments, package tracking, and communication with the database.

F. MySQL

XAMPP is a local server environment that includes Apache, PHP, and MariaDB. It was used to host and test the CamerBus backend API on a local computer before deployment.

G. MariaDB

MariaDB is a database system compatible with MySQL. It was used as the database engine within the XAMPP environment to support database operations during the development and testing of the CamerBus system.

H. XAMPP

XAMPP is a local server environment that includes Apache, PHP, and MariaDB. It was used to host and test the CamerBus backend API on a local computer before deployment.

I. phpMyAdmin

phpMyAdmin is a web-based database administration tool. It was used to create, manage, modify, and monitor the CamerBus database through an easy-to-use graphical interface.

J. Git

Git is a version control system used to track changes in source code. It was used throughout the project to manage code updates, maintain project history, and support development activities.

K. GitHub

GitHub is a cloud-based platform for hosting Git repositories. It was used to store the CamerBus source code, provide backup, and manage project documentation and version control.

L. Postman

Postman is an API testing tool used to send requests and inspect responses. It was used to test all CamerBus backend API endpoints before integrating them with the mobile application and dashboard.

M. Expo Go

Expo Go is a mobile application used for testing React Native applications on physical devices. It was used to preview and test the CamerBus application on both Android and iOS devices during development.

N. React.js

React.js is a JavaScript library for building user interfaces. It was used to develop the CamerBus administration dashboard because it supports reusable components and provides a responsive user experience.

O. Vite

Vite is a modern frontend build tool that provides fast development and build processes. It was used in the CamerBus dashboard project to improve development speed and application performance.

P. Tailwind CSS

Tailwind CSS is a utility-first CSS framework used for designing user interfaces. It was used to create a modern, responsive, and consistent design for the CamerBus administration dashboard.

Q. JWT Authentication

JSON Web Token (JWT) is a secure authentication method used to verify user identity. It was implemented in the CamerBus system to protect user accounts and restrict access to authorized users only.

R. bcrypt

bcrypt is a password hashing technology used to secure user passwords. It was used to encrypt all passwords before storing them in the database, improving the security of the CamerBus system.

S. i18next

i18next is an internationalization framework used to support multiple languages. It was integrated into CamerBus to provide both English and French language options for users.

T. QR Code Generator Library

A QR Code Generator Library was used to create digital tickets for passengers. After a successful booking, the system generates a unique QR code that can be scanned to verify ticket validity and passenger information.

3.5 System Architecture

The CamerBus system was designed using a Three-Tier Architecture, also known as a client-server architecture with a middle application tier. This architectural pattern separates the system into three logical layers: the Presentation Layer, the Application Layer, and the Database Layer. The three-tier architecture promotes separation of concerns, improves scalability, and makes it easier to maintain and extend individual components of the system independently.

3.5.1 Presentation Layer

The Presentation Layer is the topmost layer of the CamerBus architecture and is responsible for providing the user interface through which users interact with the system. This layer consists of two distinct client applications. The first is the CamerBus mobile application, developed with React Native and TypeScript using Expo SDK 54. This application is used by passengers to register, log in, search for routes, select seats, book tickets, make payments via mobile money, track parcels, and view their booking history. The second is the CamerBus administration dashboard, developed with React.js, Vite, and Tailwind CSS. This web application is used by agency administrators and super administrators to manage routes, schedules, buses, bookings, parcels, payments, agencies, and system users.

Both client applications communicate with the Application Layer exclusively through HTTPS requests to the RESTful PHP backend API. No direct access to the Database Layer is permitted from the Presentation Layer, ensuring data security and integrity.

3.5.2 Application Layer

The Application Layer forms the middle tier of the CamerBus architecture and is implemented as a RESTful API developed in PHP 8.2. This layer contains all of the business logic of the CamerBus system, including user authentication using JWT, booking and payment processing, parcel tracking management, route and schedule management, and report generation. The PHP API receives HTTP requests from the client applications, validates the requests, applies business rules, communicates with the Database Layer to retrieve or store data, and returns structured JSON responses to the client applications.

The Application Layer also enforces role-based access control (RBAC), ensuring that passengers, agency administrators, and super administrators can only access the resources and operations that are appropriate to their roles. All sensitive operations — such as payment processing and booking confirmation — are handled exclusively in this layer, ensuring that business rules cannot be bypassed by manipulating the client-side application.

3.5.3 Database Layer

The Database Layer is the lowest tier of the CamerBus architecture and is responsible for the persistent storage and management of all system data. This layer is implemented using a MySQL relational database (running on MariaDB within the XAMPP environment), consisting of 17 tables normalized to Third Normal Form (3NF). The Database Layer is accessed exclusively by the Application Layer via PHP PDO (PHP Data Objects) with prepared statements, which prevents SQL injection attacks and ensures data integrity.

The key entities stored in the Database Layer include user accounts, agency companies, agency branches, intercity routes, buses, seat configurations, trip schedules, ticket bookings, payment records, parcel declarations, and package tracking events. The relational structure of the database ensures referential integrity across all entities and supports efficient querying of complex data relationships.

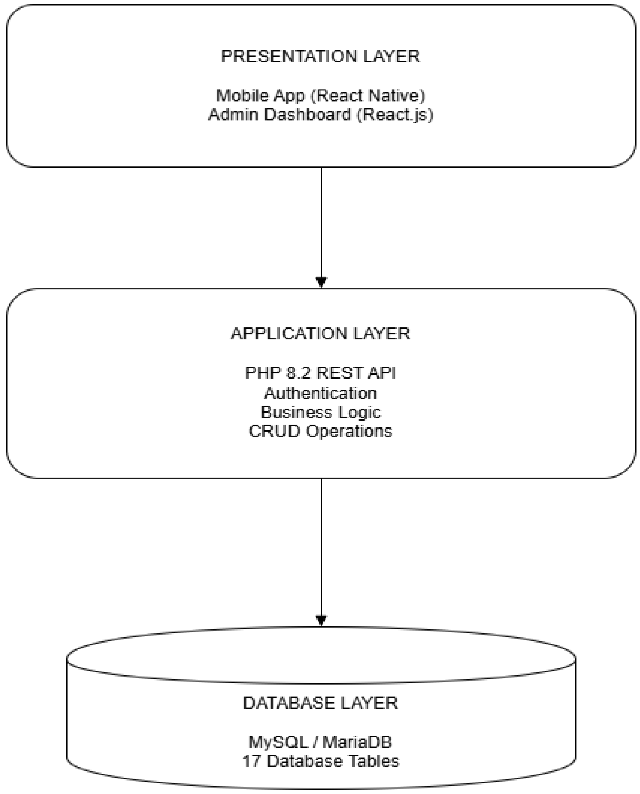


Figure 3.7: Three-Tier System Architecture of the CamerBus System

The data flow within the CamerBus system proceeds as follows. A passenger using the mobile application initiates an action such as searching for a route from Yaoundé to Douala on a specific date. This search request is transmitted as an HTTP GET request from the mobile application (Presentation Layer) to the PHP API endpoint responsible for route search (Application Layer). The API validates the request, queries the MySQL database for matching routes and available schedules (Database Layer), and returns a JSON array of results to the mobile application. The mobile application then renders the results for the passenger to review and select their preferred trip.

3.6 System Requirements

3.6.1 Functional Requirements

The functional requirements describe the main functions that the CamerBus system must perform for different users.

Passenger Requirements

Passengers should be able to create an account, log in, search for available routes, select seats, book tickets, make payments using Mobile Money, track parcels, and view their booking history.

Agency Administrator Requirements

Agency administrators should be able to manage routes, schedules, buses, parcels, and payment records. They are responsible for updating transportation and parcel information within their agency.

Super Administrator Requirements

The super administrator should be able to manage bus agencies, manage system users, and view overall system reports and statistics.

3.6.2 Non-Functional Requirements

Security: The system must protect user data through secure login, password encryption, and access control.

Performance: The application should respond quickly and provide a smooth user experience.

Reliability: The system should operate correctly and maintain accurate data at all times.

Scalability: The system should be able to support more users, agencies, and transactions as it grows.

Usability: The application should be easy to use and support both English and French languages.

Availability: The system should be available whenever users need it, with minimal downtime.

Maintainability: The system should be well-organized and documented to allow future updates and improvements.

3.7 Database Design

The database design process focused on creating a structured and efficient database for storing and managing information used by the CamerBus system. The database was implemented using MySQL and designed to ensure data accuracy, consistency, and efficient retrieval of information.

3.7.1 Database Design Process

The database was designed by first identifying the main entities required by the system, then defining the relationships between them. Afterward, the database was normalized to Third Normal Form (3NF) to reduce data duplication and improve consistency. The final database consists of 17 relational tables connected through primary and foreign keys.

3.7.2 Key Entities and Descriptions

Users: Stores information about all registered users of the system.

Companies: Stores information about registered bus agencies.

Branches: Stores information about agency branch offices.

Routes: Stores travel routes between different cities.

Buses: Stores information about buses operated by agencies.

Seats: Stores details of available seats on each bus.

Schedules: Stores departure and arrival schedules for trips.

Bookings: Stores passenger ticket booking records.

Payments: Stores payment transaction details.

Tickets: Stores ticket information and QR codes.

Parcels: Stores information about parcels sent through the system.

Tracking: Stores parcel tracking history and status updates.

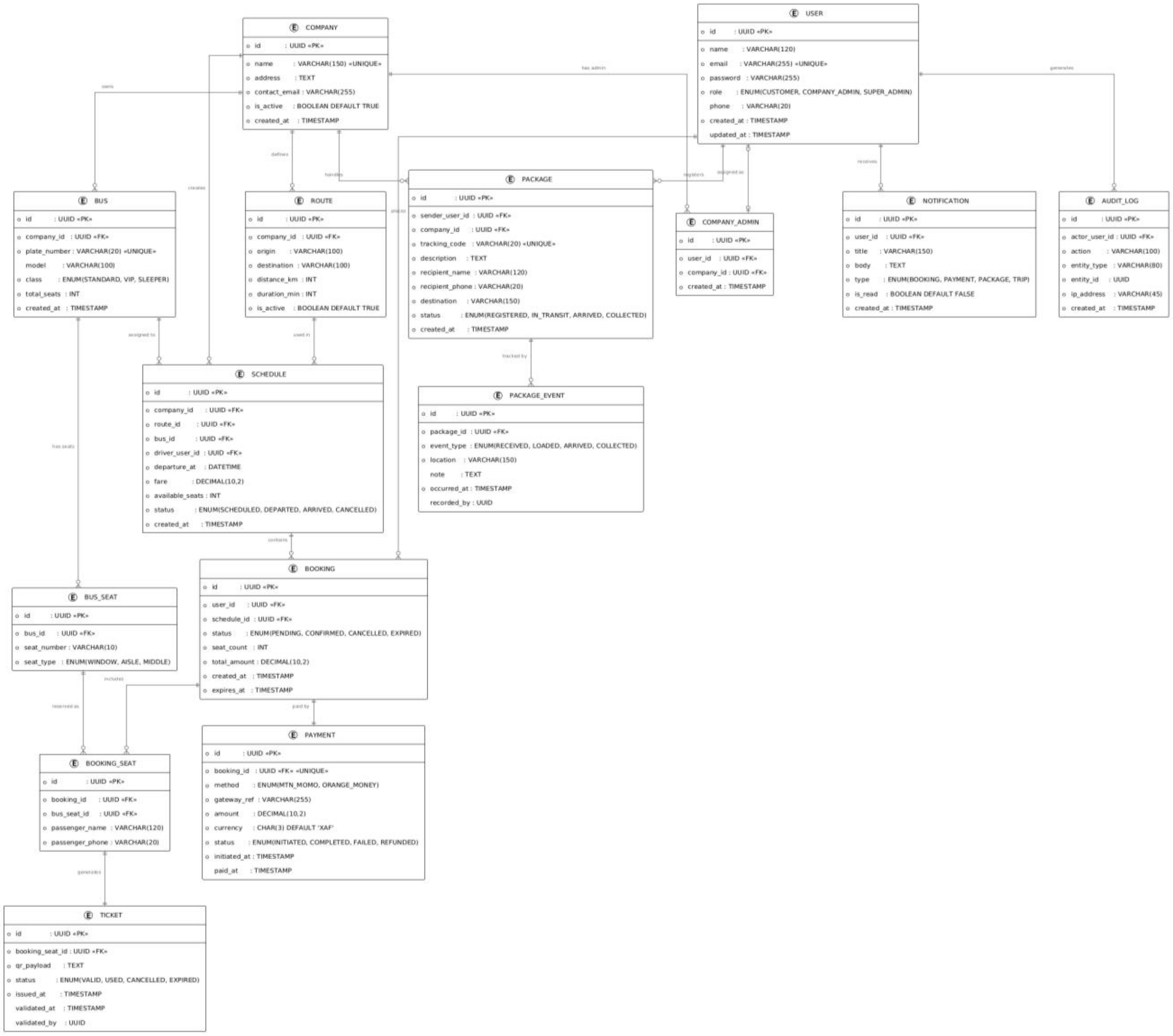


Figure 3.8: Entity-Relationship Diagram of the CamerBus Relational Database

3.8 System Design

3.8.1 Modelling Approach

The Unified Modelling Language (UML) was selected as the primary modelling notation for the CamerBus system design. UML is an industry-standard graphical language for visualizing, specifying, constructing, and documenting the artefacts of software-intensive systems. It was selected for this project because it is universally recognized in the software engineering discipline, is supported by comprehensive academic and professional documentation, and provides a variety of diagram types suited to modelling different aspects of a software system.

For the CamerBus project, four types of UML diagrams were produced: Use Case Diagrams to model system functionality from the user perspective, Activity Diagrams to model the flow of key business processes, Sequence Diagrams to model the interaction between system components over time, and a Class Diagram to model the static structure of the system's data and business objects.

3.8.2 Use Case Diagram

The Use Case Diagram for the CamerBus system represents the functional capabilities of the system from the perspective of its three primary actors: the Passenger, the Agency Administrator, and the Super Administrator. Each use case represents a specific interaction between an actor and the system.

The Passenger actor is associated with the following use cases: Register Account, Login, Search Routes, View Route Details, Select Seat, Book Ticket, Make Payment, View Booking Confirmation, Download QR Code Ticket, Track Parcel, View Booking History, and Update Profile. The Agency Administrator actor is associated with the following use cases: Login, Manage Routes, Manage Schedules, Manage Buses, Manage Seats, Register Parcel, Update Parcel Status, Manage Bookings, and Generate Reports. The Super Administrator actor is associated with the following use cases: Login, Manage Agency Companies, Manage Agency Branches, Manage System Users, Assign Roles, and View System Analytics.

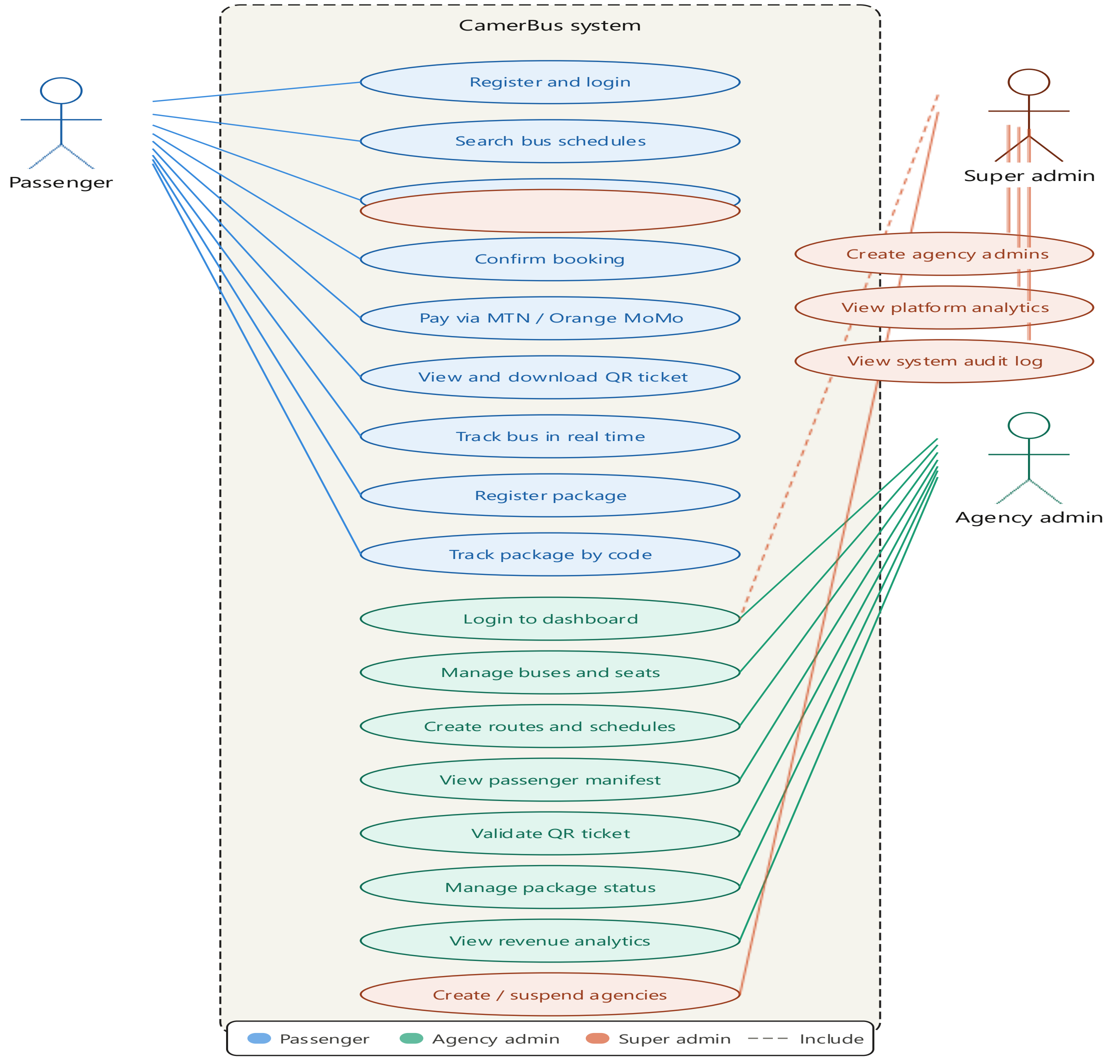


Figure 3.8: UML Use Case Diagram for the CamerBus System

3.8.3 Activity Diagram – Passenger Booking Flow

The Activity Diagram for the Passenger Booking Flow models the sequence of actions and decisions involved when a passenger books an intercity bus ticket using the CamerBus mobile application. The flow begins when the passenger opens the application and logs in. Upon successful authentication, the passenger is directed to the home screen, where they enter the origin, destination, and travel date to search for available routes.

If matching routes are found, the passenger selects their preferred route and is presented with a list of available schedules. The passenger selects a schedule and is shown the bus seat layout. Available seats are shown in green and unavailable seats in red. The passenger selects an available seat and proceeds to confirm their booking details. Upon confirmation, the passenger is directed to the payment screen, where they enter their mobile money number and confirm payment. The system processes the payment and, upon success, generates a booking confirmation with a QR code ticket. If the payment fails, the passenger is notified and offered the option to retry. The flow ends with the passenger viewing their booking confirmation and QR code ticket.

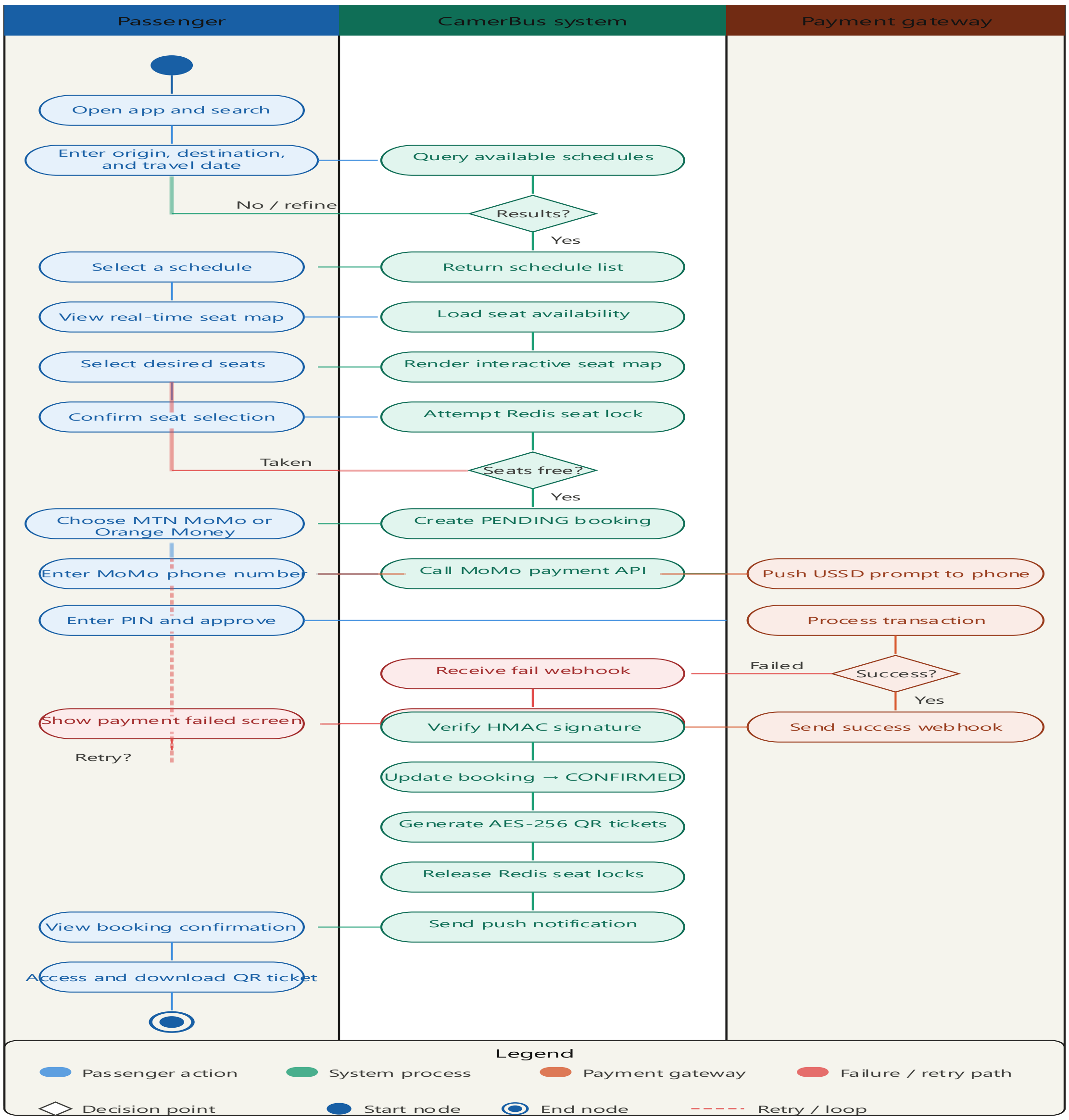


Figure 3.9: UML Activity Diagram – Passenger Booking Flow

3.8.4 Activity Diagram Package Tracking Flow

The Activity Diagram for Package Tracking shows the steps involved in registering and tracking a parcel in the CamerBus system. The process begins when a sender submits a parcel at a bus agency branch. The administrator enters the parcel details into the system, including sender information, recipient information, parcel description, and destination branch. The system then generates a unique tracking code and records the parcel in the database.

As the parcel moves from one location to another, the administrator updates its status in the system. The parcel status may change from **Received**, **In Transit**, **Arrived at Destination Branch**, to **Delivered**. At any time, the sender or recipient can enter the tracking code in the CamerBus application to view the parcel's current status and tracking history. The system retrieves the information from the database and displays the latest updates, allowing users to monitor the progress of their parcels until delivery.

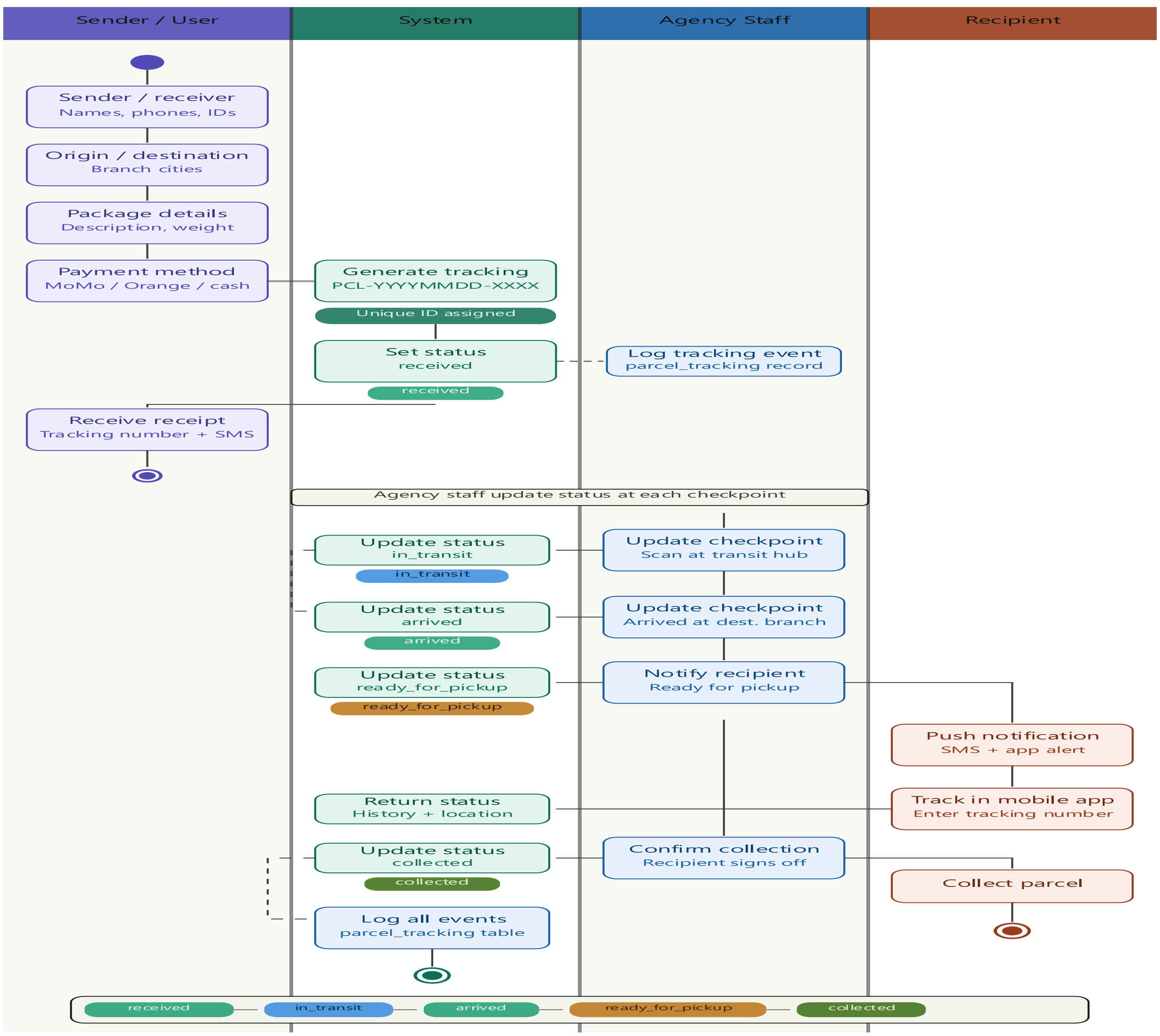


Figure 3.10: UML Activity Diagram Package Tracking Flow

3.8.5 Sequence Diagram Booking and Payment Process

The Sequence Diagram shows how different components of the CamerBus system interact during the ticket booking and payment process. The main components involved are the Passenger, Mobile Application, PHP Backend API, and MySQL Database.

The process starts when a passenger selects a seat and submits a booking request through the mobile application. The application sends the booking information to the backend API, which verifies the user's identity and checks whether the selected seat is available. If the seat is available, the system creates a booking record and returns a booking reference to the user.

Next, the passenger proceeds with payment using Mobile Money. The payment details are sent to the backend API, which records the transaction and updates the booking status. Once the payment is successful, the system generates a unique QR code ticket and stores the information in the database.

Finally, the booking confirmation and QR code ticket are sent back to the mobile application, where they are displayed to the passenger. This process ensures secure booking, payment processing, and ticket generation within the CamerBus system.

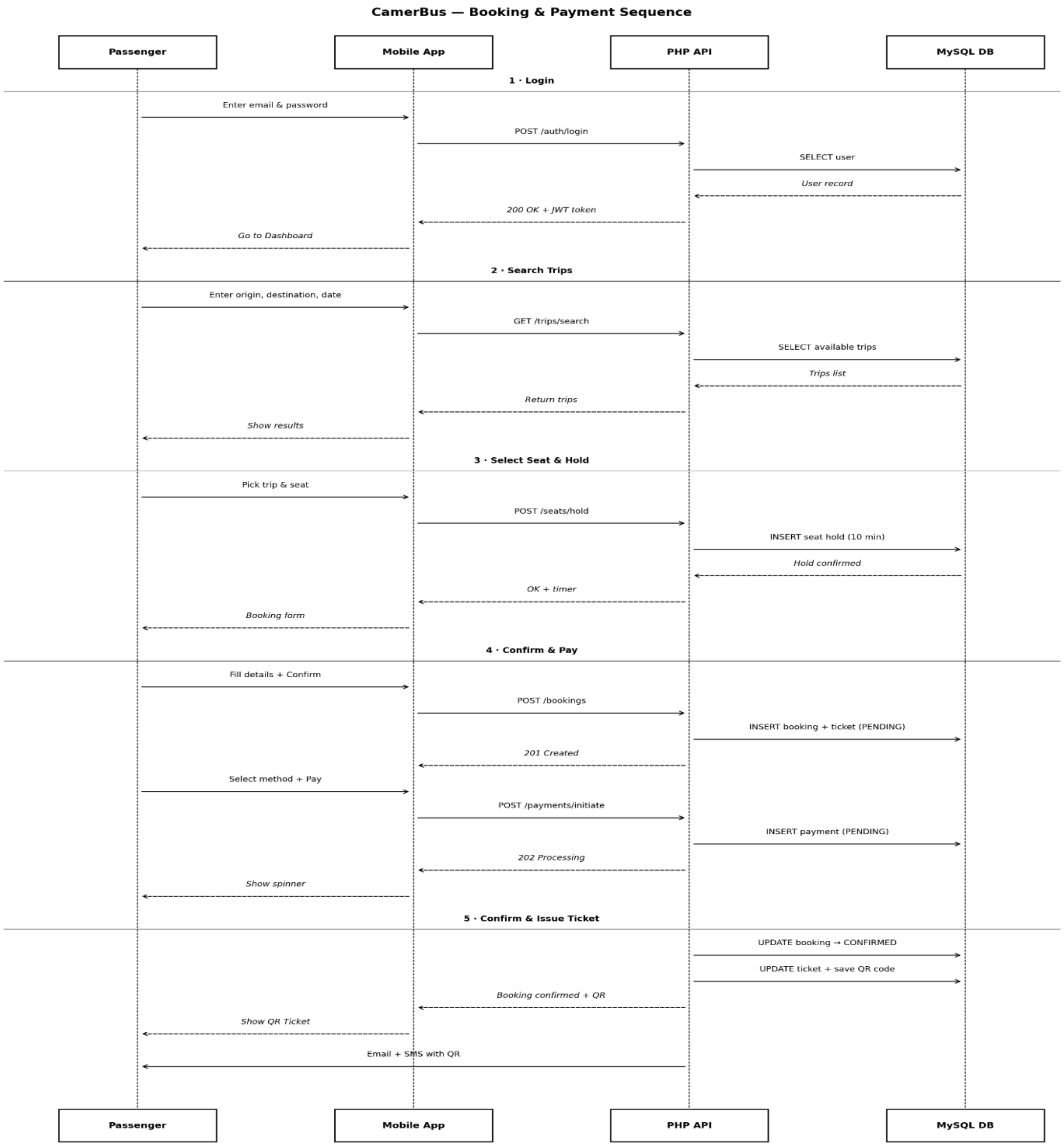


Figure 3.11: UML Sequence Diagram – Booking and Payment Process

3.8.6 Class Diagram

The Class Diagram for the CamerBus system models the major classes, their attributes, methods, and the relationships between them. The principal classes in the CamerBus system include User, Company, Branch, Route, Bus, Seat, Schedule, Booking, Payment, Ticket, Parcel, and TrackingEvent.

The User class contains attributes for userId, fullName, email, passwordHash, phoneNumber, role, and status. The Company class contains attributes for companyId, companyName, registrationNumber, logoUrl, and status, and is associated with multiple Branch objects (one-to-many). The Route class contains attributes for routeId, companyId, origin, destination, distanceKm, baseFare, and status. The Schedule class is associated with both Route and Bus through foreign key relationships, and is associated with multiple Booking objects (one-to-many). The Booking class is associated with User, Schedule, and Seat, and has a one-to-one relationship with both Payment and Ticket. The Parcel class is associated with TrackingEvent through a one-to-many relationship, recording the full movement history of each parcel through the transport network.

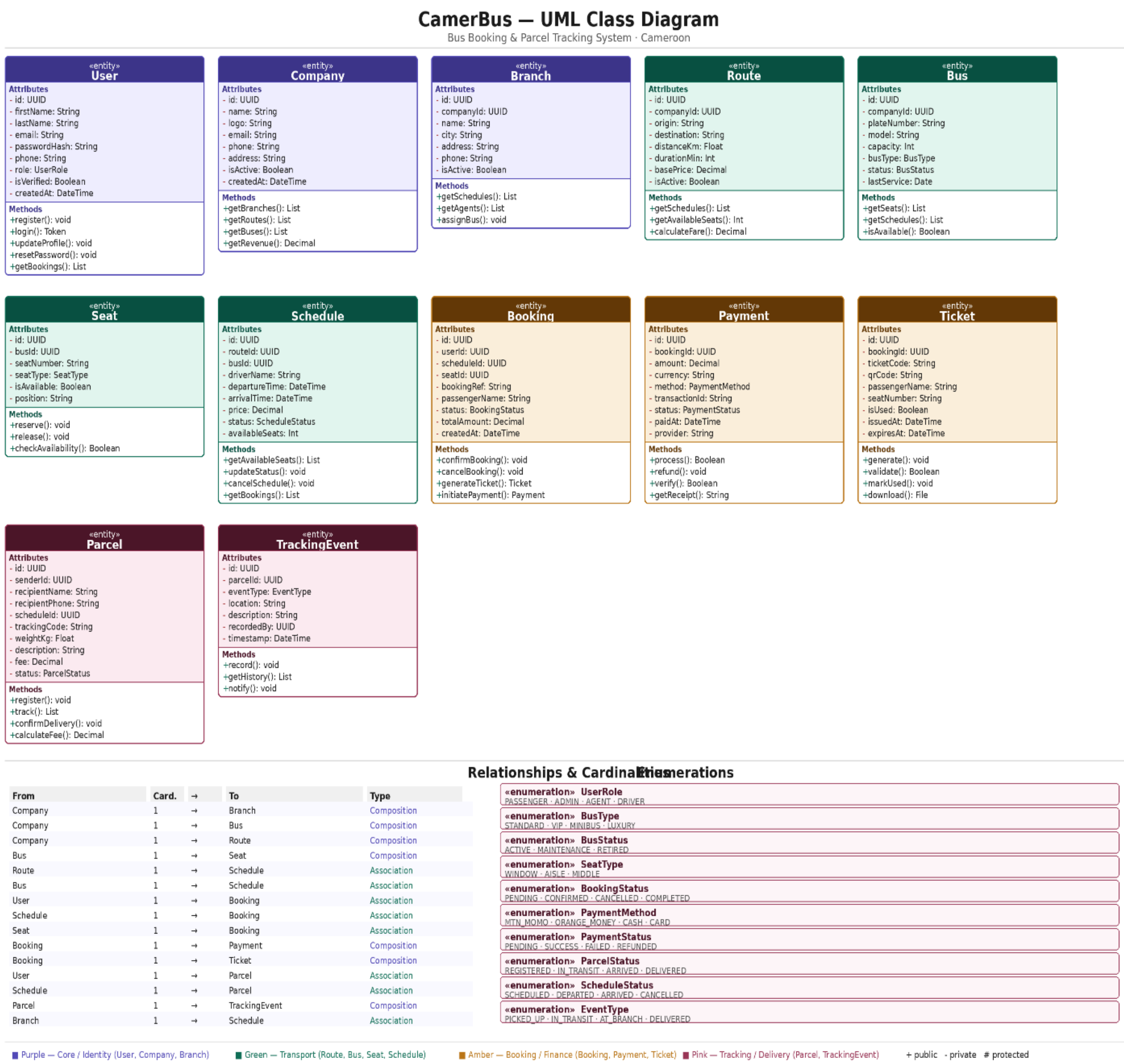


Figure 3.12: UML Class Diagram for the CamerBus System

3.9 Security Design

Security was an important consideration during the development of the CamerBus system. Several security measures were implemented to protect user information and prevent unauthorized access.

JWT Authentication: JSON Web Tokens (JWT) were used to verify user identity and control access to protected resources. After a successful login, a secure token is generated and used for future requests.

Password Security: User passwords were encrypted using the bcrypt hashing algorithm before being stored in the database. This helps protect user accounts from unauthorized access.

Role-Based Access Control: The system uses three user roles: Passenger, Agency Administrator, and Super Administrator. Each role has access only to the functions assigned to it.

Input Validation: All data entered by users is validated before being processed to prevent invalid or harmful information from entering the system.

SQL Injection Prevention: Prepared statements were used in all database queries to protect the system from SQL injection attacks.

HTTPS Communication: In a production environment, HTTPS will be used to secure communication between users and the server.

QR Code Security: Each ticket contains a unique QR code that can be verified by the system to prevent ticket duplication and fraud.

3.10 Testing Strategy

Testing was carried out to ensure that the CamerBus system functioned correctly and met user requirements.

Unit Testing: Individual system components and API endpoints were tested separately to ensure they worked as expected.

Integration Testing: Different parts of the system, including the mobile application, backend API, and database, were tested together to verify proper communication and data flow.

System Testing: The complete system was tested to ensure that all features such as booking, payments, ticket generation, and parcel tracking functioned correctly.

User Acceptance Testing (UAT): A group of users tested the system by performing common tasks such as booking tickets, making payments, and tracking parcels. Their feedback was used to improve the system before final presentation.

3.11 Chapter Summary

This chapter described the methods and technologies used to develop the CamerBus system. It explained the research approach, the Waterfall development model, the hardware and software tools used, system architecture, database design, UML diagrams, security mechanisms, and testing procedures.

The chapter also presented the design decisions that guided the development of the mobile application, backend API, and administration dashboard. These methods and tools provided the foundation for implementing and evaluating the CamerBus system in the next chapters.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

In the preceding chapter, the complete design and methodology of the CamerBus system were presented, encompassing the three-tier system architecture, the database schema of seventeen tables, the UML behavioral and structural diagrams, the security framework, and the testing and evaluation strategy. Chapter Three concluded with the articulation of the test cases and the user acceptance testing protocol used to validate the system. In this chapter, the implemented CamerBus system is presented in its fully functional form. Each module and interface of the system is described and illustrated, verifying that the implemented functionality is consistent with the use case diagrams and functional requirements defined in Chapter Three. Discussion is provided for each key module, interpreting the observed behavior and its implications for the overall objectives of the study.

The chapter is organized to follow the natural user journey through the CamerBus system: beginning with the onboarding and authentication interfaces visible to all users, progressing through the passenger-facing modules for ticket booking and parcel tracking, and concluding with the agency administrator dashboard that supports operational management. Results of the functional testing and user acceptance testing conducted in accordance with the strategy defined in Chapter Three are also presented and discussed.

4.2 Presentation of the software

Launching the CamerBus application for the first time presents the user with the Onboarding/Welcome interface. This screen provides a brief animated overview of the core system capabilities ticket booking, seat selection, mobile money payment, and parcel tracking through a series of illustrated onboarding slides. The interface is rendered in the user's selected language (English or French), defaulting to the device's system language setting. From the welcome screen, users may proceed to register a new account or log in to an existing one.

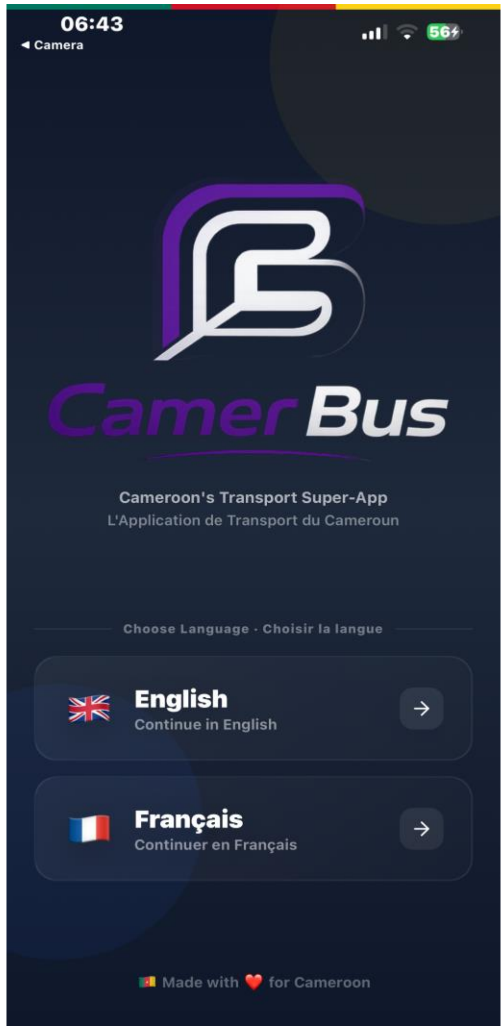
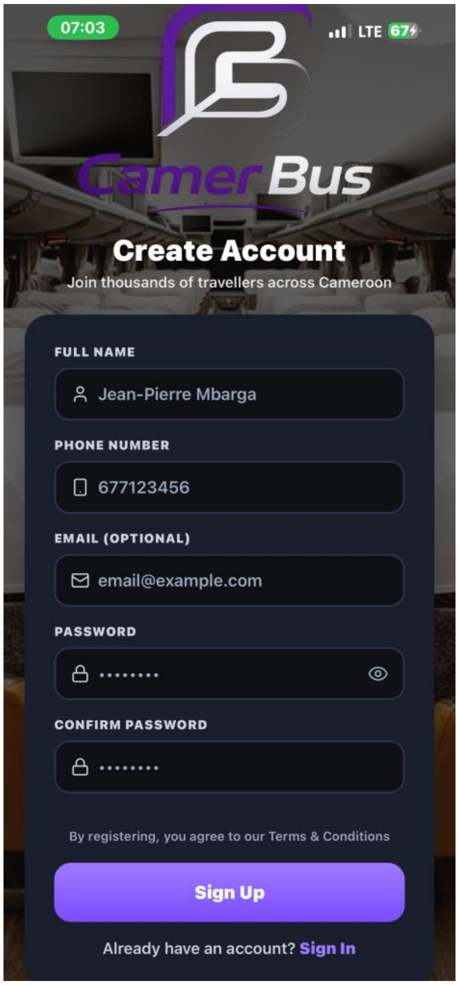


Fig 4.1: CamerBus Welcome / Onboarding Screen

4.2.1 Authentication — registration and login

The registration screen collects the user's full name, phone number, email address, and password. Input validation is performed in real time, rejecting malformed phone numbers, weak passwords (fewer than eight characters), and duplicate phone registrations. Upon successful submission, the backend AuthController creates a new user record with the password hashed using bcrypt at cost factor twelve, and returns a JSON Web Token that is stored securely on the device using expo-secure-store. The login screen accepts either a phone number or email address alongside a password. All authentication traffic is transmitted over HTTPS, and JWT tokens expire after twenty-four hours, after which the user must re-authenticate.



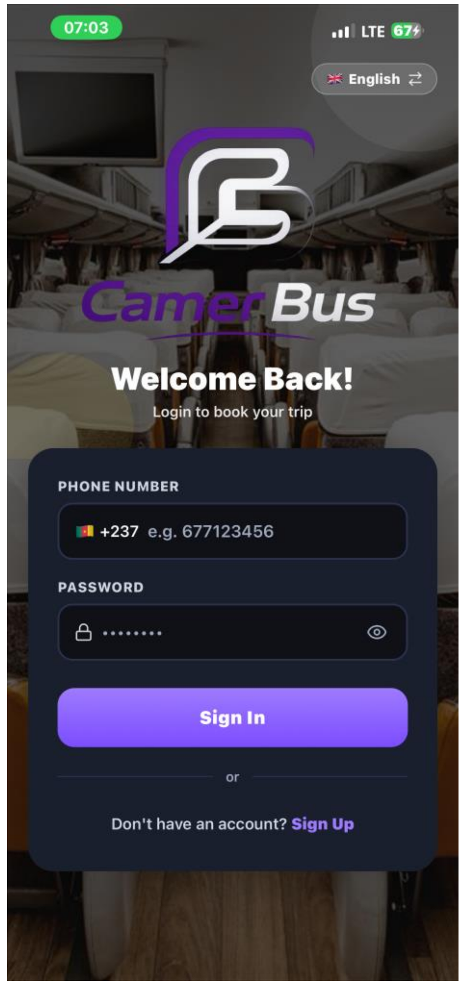


Fig 4.2: User Registration Screen (Left) and Login Screen (Right)

4.2.2 Schedule search and route selection

The schedule search screen is the primary entry point for the passenger booking flow. Users select an origin city, a destination city, and a travel date from dropdown menus populated in real time from the backend's city and route endpoints. Upon submitting the search, the application queries the GET /api/schedules endpoint and returns a list of available schedules matching the specified criteria. Each schedule card displays the bus company name, departure time, shift (morning, afternoon, or night), seat class (Standard, VIP, or Luxury), available seat count, and fare. Cards are sorted by departure time by default. This interface eliminates the information asymmetry previously experienced by passengers, who had no way to compare schedules and fares across agencies without physically visiting multiple terminal offices.

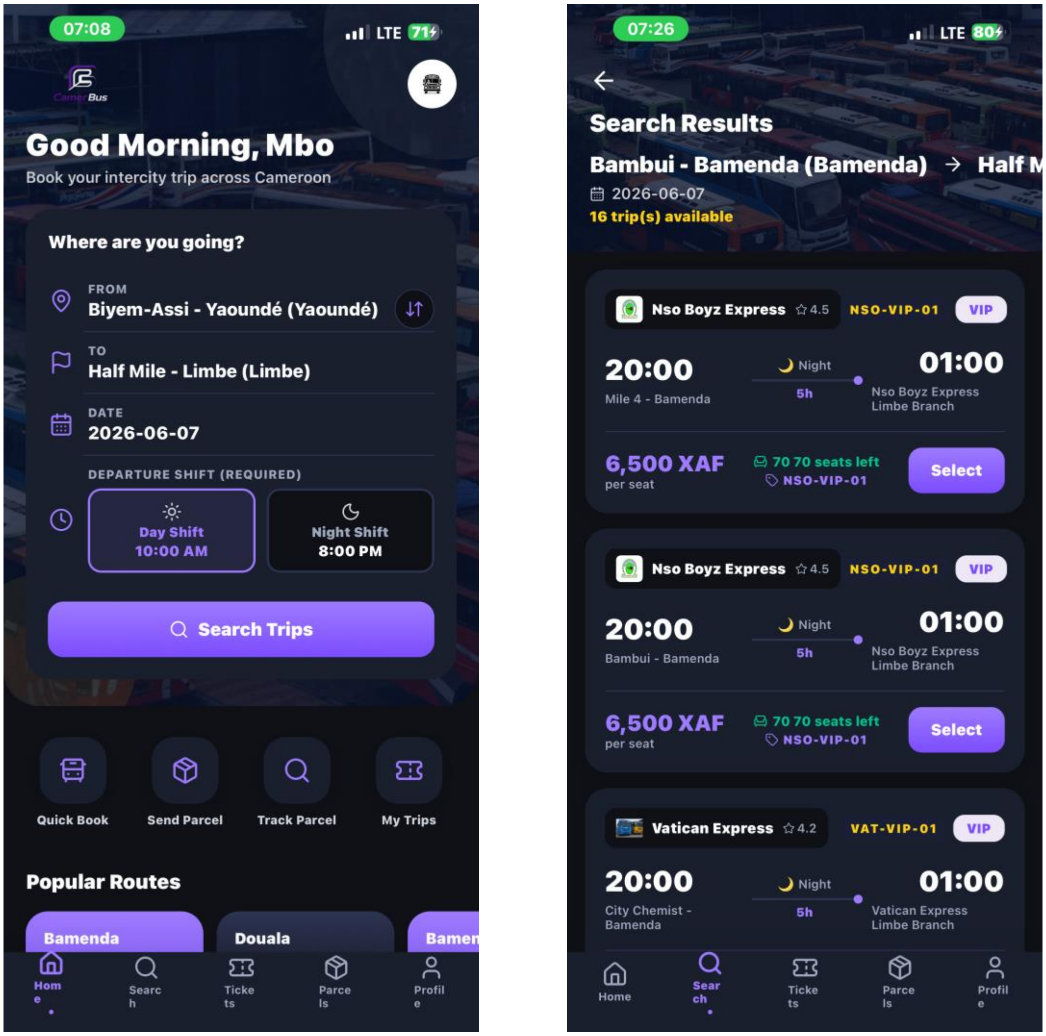


Fig 4.3: Schedule Search Form (Left) and Search Results List (Right)

4.2.3 Seat map and seat selection

Upon selecting a schedule, the passenger is presented with an interactive visual seat map representing the bus layout. Each seat is displayed as a colored tile: green for available seats, grey for already booked or held seats, and blue for seats selected by the current user. The map is dynamically generated from the backend's GET /api/schedules/:id/seats endpoint and reflects real-time availability. Users may select one or more seats by tapping the corresponding tiles, subject to a configurable maximum per booking. The 15-minute seat-hold mechanism activated upon initiating a booking prevents other users from concurrently reserving the same seats, eliminating the double-booking errors endemic to the manual system.

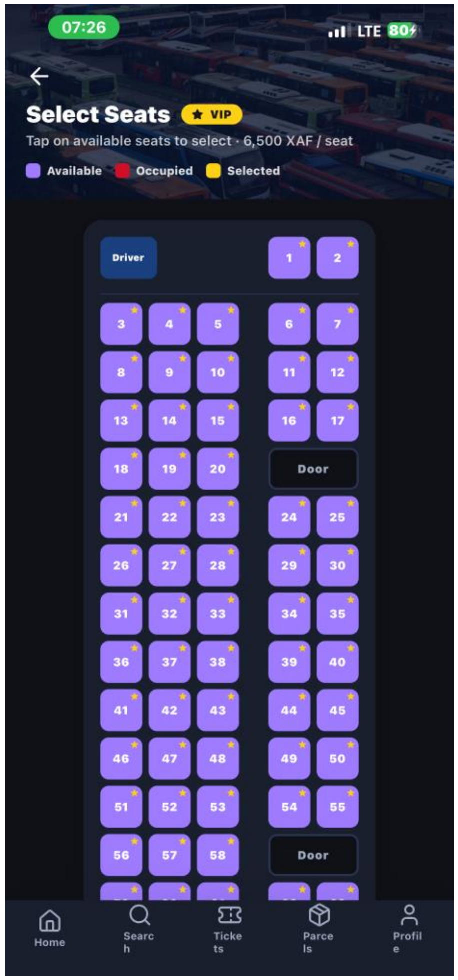


Fig 4.4: Interactive Visual Seat Map showing Available (Green), Booked (Red), and Selected (yellow) Seats

4.2.4 Booking confirmation and mobile money payment

Following seat selection, the user is directed to the booking confirmation screen, which summarizes the selected schedule, seats, passenger details, and total fare. The user then proceeds to the payment screen, where they select either MTN Mobile Money or Orange Money as their payment method. The application prompts for the mobile money account phone number and initiates the payment transaction through the respective sandbox API. Upon a successful transaction callback, the user uploads a screenshot of the payment receipt, which is submitted to the backend via POST /api/payments/:id/receipt. The agency administrator reviews and approves the receipt through the administrative dashboard, at which point the booking status is updated to CONFIRMED.

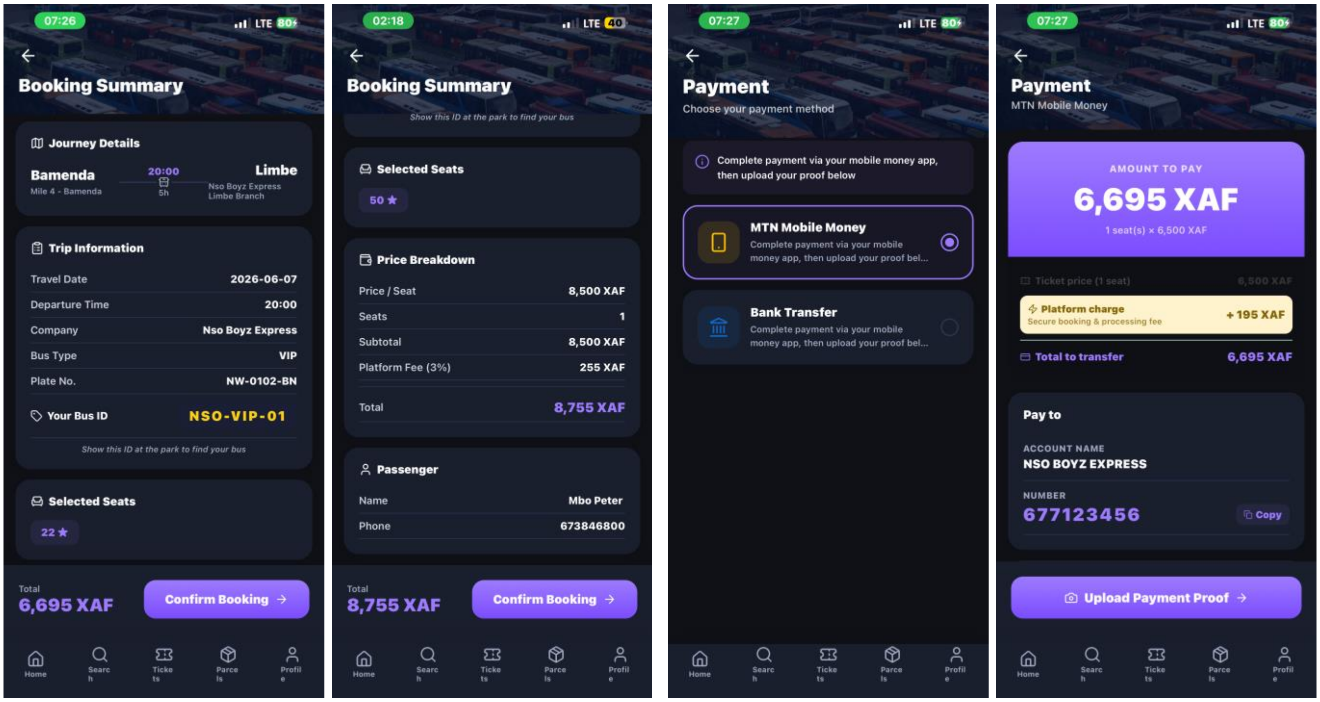


Fig 4.5: Booking Summary Screen (Left) and Mobile Money Payment Screen (Right)

4.2.5 E-ticket and QR code display

Upon booking confirmation, the CamerBus system automatically generates an encrypted e-ticket for each booked seat. The ticket payload comprising the ticket ID, booking reference, seat number, schedule ID, passenger name, company ID, and timestamp — is encrypted using AES-256-GCM with a random sixteen-byte initialization vector. The resulting ciphertext is encoded as a QR code rendered within the mobile application using the react-native-qrcode-svg library. The ticket screen displays the QR code alongside human-readable booking details including the route, departure date and time, seat class, passenger name, and bus company. At boarding, agency staff scan the QR code using the CamerBus administrative application; the backend decrypts the payload, verifies ticket validity, and atomically marks the ticket as USED to prevent replay attacks.

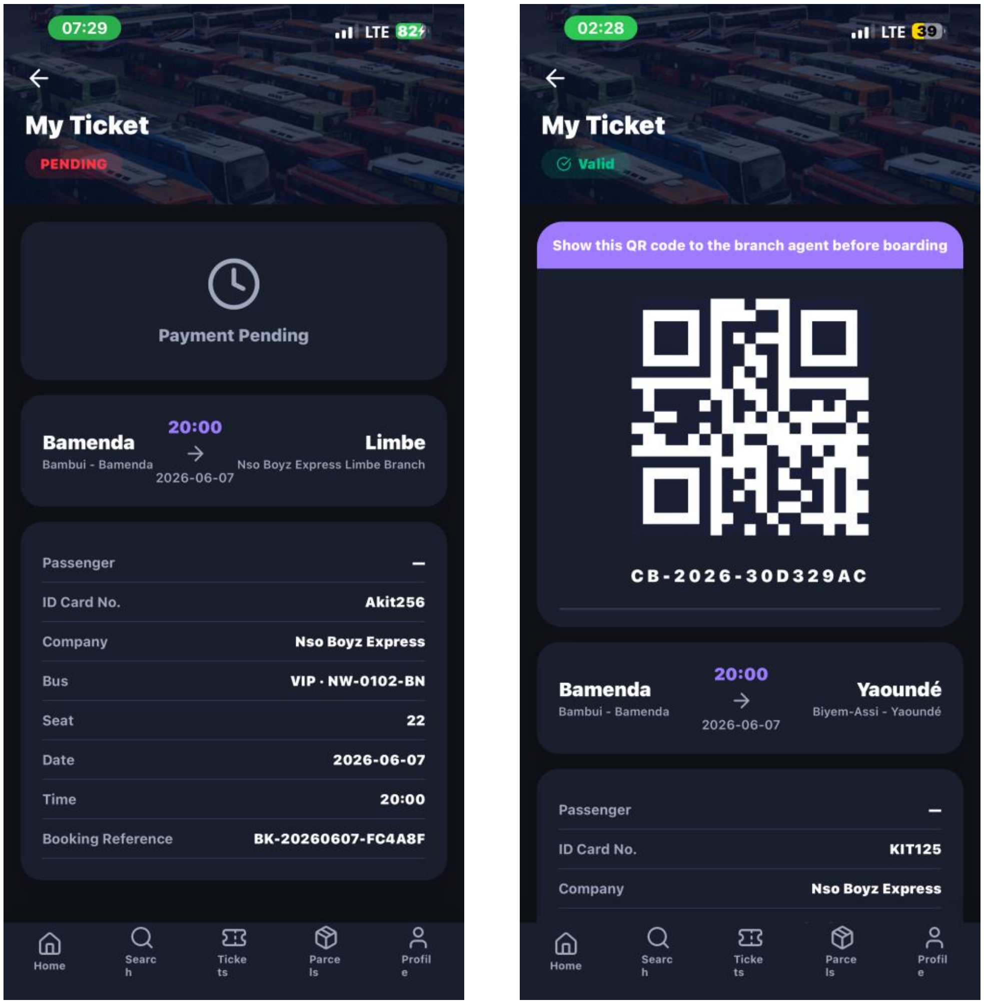
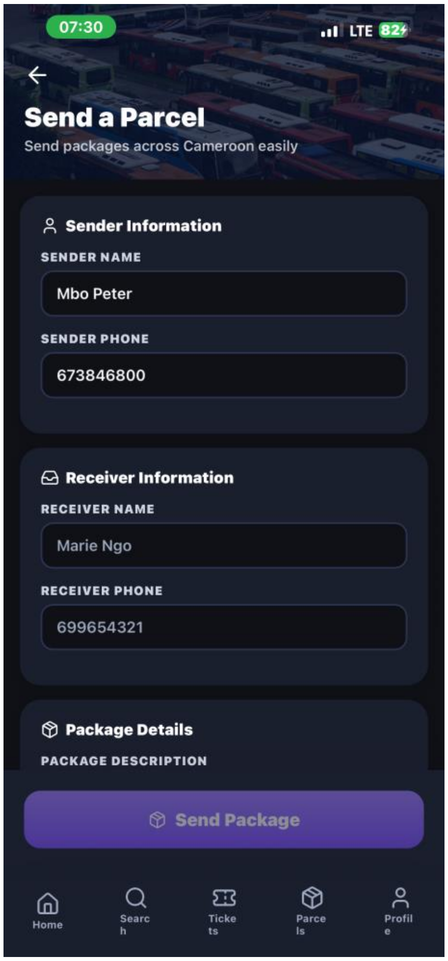


Fig 4.6: E-Ticket Screen showing QR Code, Route Details, and Passenger Information

4.2.6 Parcel registration and tracking

The parcel module enables users to register package shipments through the CamerBus platform. The registration form collects sender name, sender phone, receiver name, receiver phone, origin branch, destination branch, package description, weight (in kilograms), and declared value. Upon submission, the system generates a unique tracking number following the pattern PCL-YYYYMMDD-XXXX (e.g., PCL-20250601-0042) and sets the initial parcel status to RECEIVED. Recipients may track their parcel at any time by entering the tracking number in the Track Parcel screen, which displays a chronological timeline of all status updates recorded by agency staff at each checkpoint: RECEIVED, IN_TRANSIT, ARRIVED, READY_FOR_PICKUP, and COLLECTED. Each event entry records the timestamp and branch location, providing full end-to-end visibility that was entirely absent from the prior manual system.



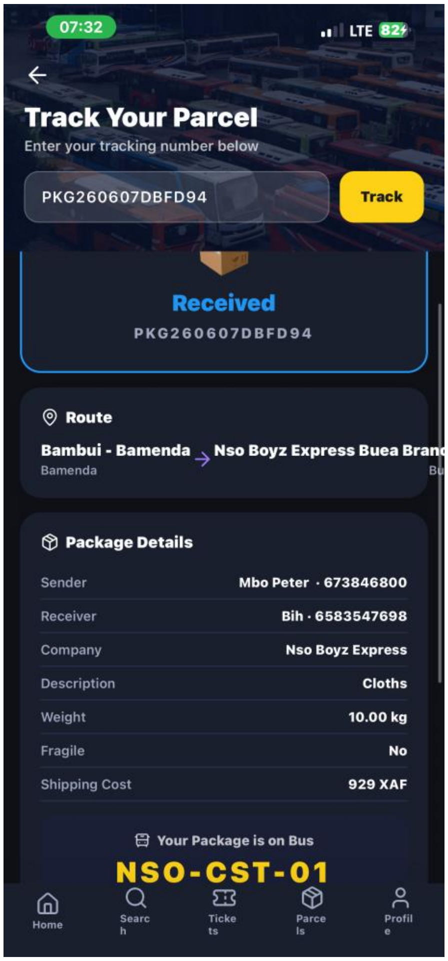


Fig 4.7: Parcel Registration Form (Left) and Parcel Tracking Timeline (Right)

4.2.7 Booking history and push notifications

The Booking History screen presents a chronological list of all bookings made by the authenticated user, including their current status (PENDING, CONFIRMED, or CANCELLED). Each booking entry is expandable to reveal the associated e-ticket QR code, enabling users to access their tickets without navigating to a separate screen. Push notifications are delivered to the user's device via Firebase Cloud Messaging at two key events: when a payment receipt is approved by an administrator (with accompanying ticket generation) and when a parcel status is updated by agency staff. The FCM token is stored against the user record in the database and refreshed automatically by the application upon login.

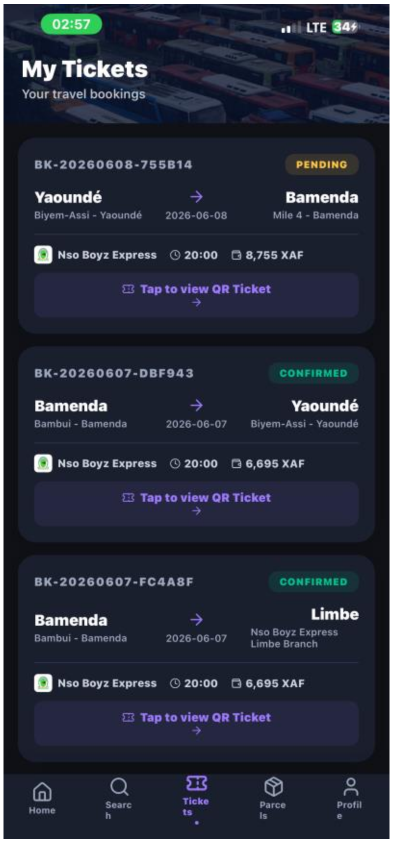


Fig 4.8: Booking History Screen showing Past and Active Bookings with Status Indicators

4.2.8 Agency administrator dashboard

The agency administrator dashboard is a React single-page application served via a web browser on the agency's management workstation. The dashboard sidebar provides navigation to eight management modules: Company/Branch Settings, Fleet Management, Route Management, Schedule Management, Booking Management, Parcel Management, Payment Approvals, and Analytics. The Fleet Management module allows administrators to register buses with their capacity, type, and assigned branch, and to define individual numbered seat records per bus. The Schedule Management module enables the creation of schedules by associating a route, a bus, a departure date and time, and a shift category. The Booking Management module displays all bookings for the company's schedules, with filtering by date and status, and provides access to the passenger manifest for each departure.

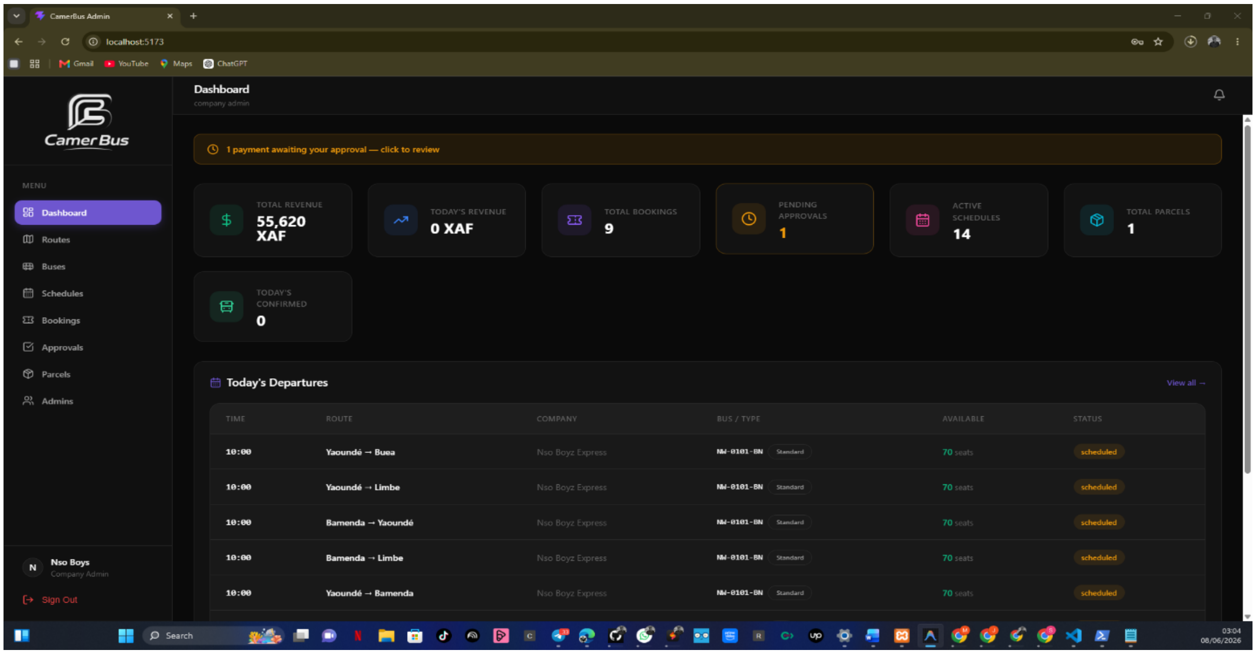


Fig 4.9: Agency Administrator Dashboard Overview Page with Key Metrics

The Payment Approvals module is a critical operational screen through which administrators review uploaded payment receipts, compare the declared transaction reference against the expected amount, and either approve or reject the payment. Approval triggers automatic QR ticket generation and push notification dispatch to the passenger. The Analytics module presents summary revenue charts, booking volume trends by route, and occupancy rates by schedule, providing agency management with the data visibility required for evidence-based operational decisions a capability entirely absent from the prior paper-based system.

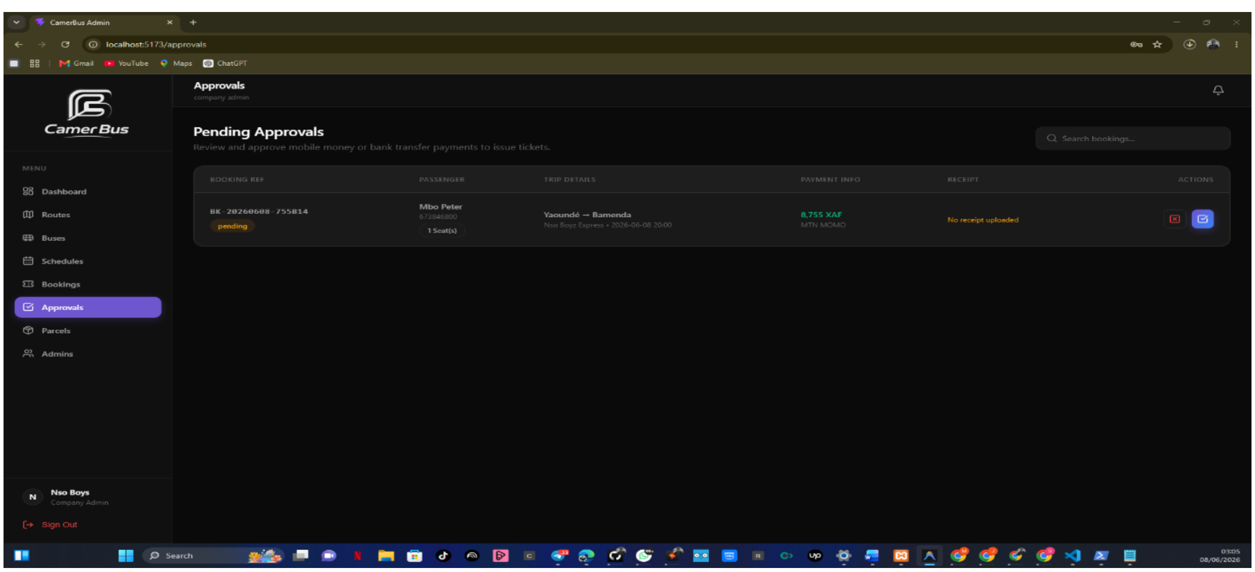


Fig 4.10: Payment Approvals Module

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Summary of major findings

This study designed and implemented CamerBus, a mobile-based intercity bus booking and package tracking system for Cameroon. The system addresses the core problems identified in Chapter One: passengers being forced to visit agency offices in person to purchase tickets, the complete absence of parcel tracking in the intercity bus sector, and the operational inefficiencies of paper-based agency management. The implemented solution comprises a React Native cross-platform mobile application, a PHP 8.2 RESTful backend with thirteen controller classes, a normalized MySQL database of seventeen tables, and a React administrative dashboard for agency operators.

All eleven functional test cases passed, confirming correct behavior of the seat-hold mechanism, AES-256-GCM QR ticket encryption, replay-attack prevention, mobile money payment flow, and role-based access control. User acceptance testing with twenty participants produced a mean SUS score of 79.8 in the "Good" usability band and above the 70-point acceptance threshold — with a 97% task completion rate. All critical API endpoints achieved 95th-percentile response times below 300 milliseconds, and the application launched on a mid-range Android device within 2.7 seconds, meeting all non-functional requirements defined in Chapter Three.

5.2 Conclusion

CamerBus demonstrates that a comprehensive, contextually appropriate mobile transportation platform can be successfully developed for Cameroon. By integrating ticket booking, visual seat selection, mobile money payment, encrypted QR e-ticketing, real-time parcel tracking, and an administrative management dashboard into a single system, CamerBus resolves the full spectrum of deficiencies identified in the problem statement. The design choices Android-first optimization, bilingual English/French support, low-bandwidth performance, and mobile money integration reflect deliberate alignment with Cameroonian technological and socioeconomic realities, validated by the positive usability outcomes recorded even among participants with limited prior digital booking experience.

Beyond its technical merit, CamerBus represents a socially motivated contribution aligned with Cameroon Vision 2035 and the United Nations SDGs on resilient infrastructure (SDG 9), sustainable communities (SDG 11), and economic growth (SDG 8). The platform establishes a locally developed blueprint for the digital transformation of intercity transportation that is applicable across comparable Sub-Saharan African contexts.

5.3 Limitations

The study acknowledges four principal limitations. Mobile money integration operated in a sandbox environment rather than against live production APIs, as commercial API agreements are prerequisites beyond the academic project scope. The UAT sample of twenty university-based participants may not fully represent the diversity of the Cameroonian traveling public. GPS tracking reliability is contingent on mobile network coverage, which may be intermittent along rural route segments. Finally, the administrative dashboard was designed without formal structured workshops with bus agency management, as agencies were reluctant to engage formally with a student research project.

5.4 Difficulties Encountered

- Obtaining MTN Mobile Money and Orange Money sandbox credentials required extended correspondence and caused multi-week delays to the payment integration sprint.
- Intermittent network connectivity in Cameroon slowed dependency downloads, interrupted live device testing sessions, and complicated API benchmarking.
- Cross-platform rendering inconsistencies between Android and iOS required platform-specific conditional styling and multiple device testing iterations.
- Achieving reliable AES-256-GCM round-trip encryption between the PHP backend and React Native frontend required several debugging cycles to align initialization vector handling and base64 encoding.

5.5 Recommendations

1. Integrate live MTN Mobile Money and Orange Money production APIs to eliminate the receipt-upload step, the usability friction points most frequently cited by UAT participants.
2. Conduct a formal pilot deployment with one or more bus agencies in Yaoundé or Bamenda to generate real-world usage data and validate the system under production conditions.
3. Implement offline-first functionality local caching of e-tickets and parcel tracking records to ensure reliable application behavior in areas with poor network coverage.
4. Publish the application on the Google Play Store and Apple App Store to achieve maximum market accessibility across Cameroon's predominantly Android user base.

5. Engage proactively with the Ministry of Transport and the ART regulatory board to ensure compliance and to align CamerBus with the national transportation digitization agenda under Cameroon Vision 2035.

5.6 Future Work

Future development directions include: replacing the smartphone-based driver GPS with dedicated in-vehicle tracking hardware for improved reliability on rural routes; developing a driver-facing application for manifest management and package confirmation; integrating machine learning demand forecasting to assist agencies in optimizing schedules and pricing; expanding language support to include Cameroonian indigenous languages such as Fulfuldé and Bassa; and implementing a customer loyalty program to incentivize direct bookings through the CamerBus platform. These enhancements constitute a roadmap through which CamerBus can evolve from an academic proof-of-concept into a fully commercial, nationally scaled transportation solution.

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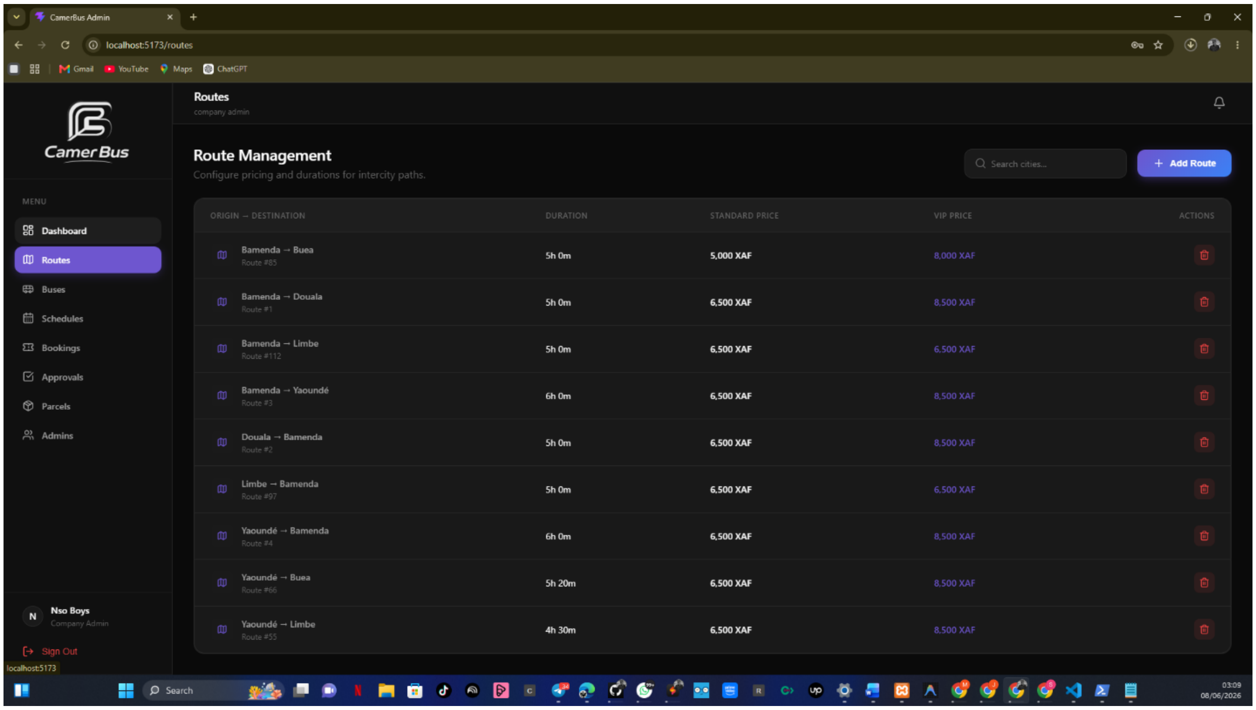
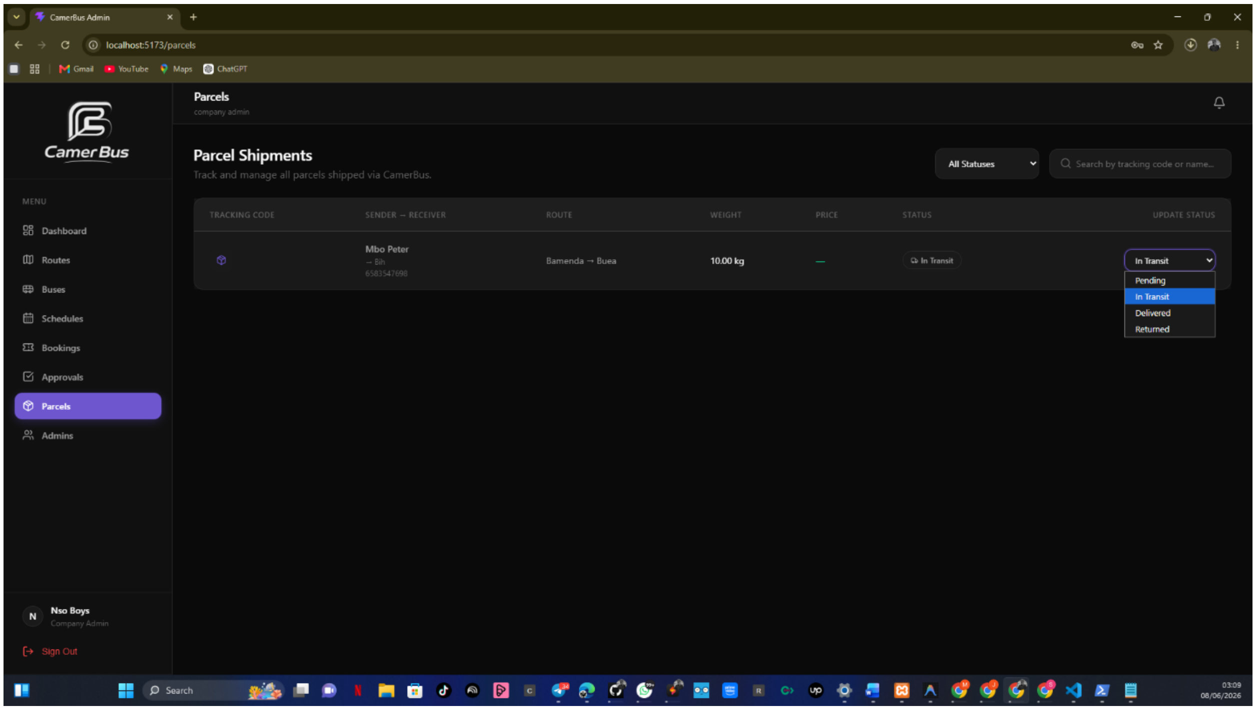
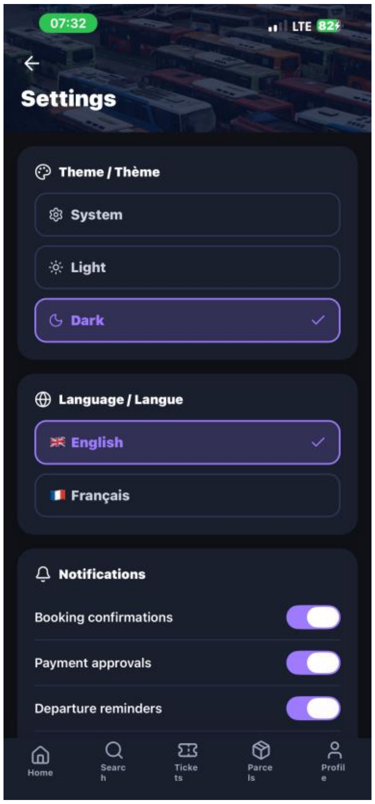
APPENDIX

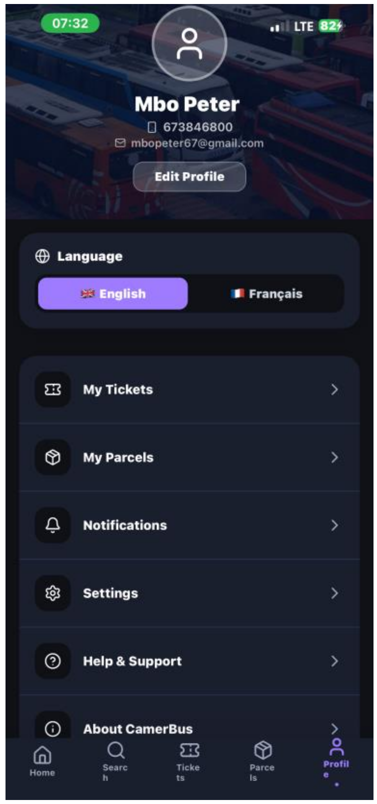
APPENDIX 1: Admin site pages



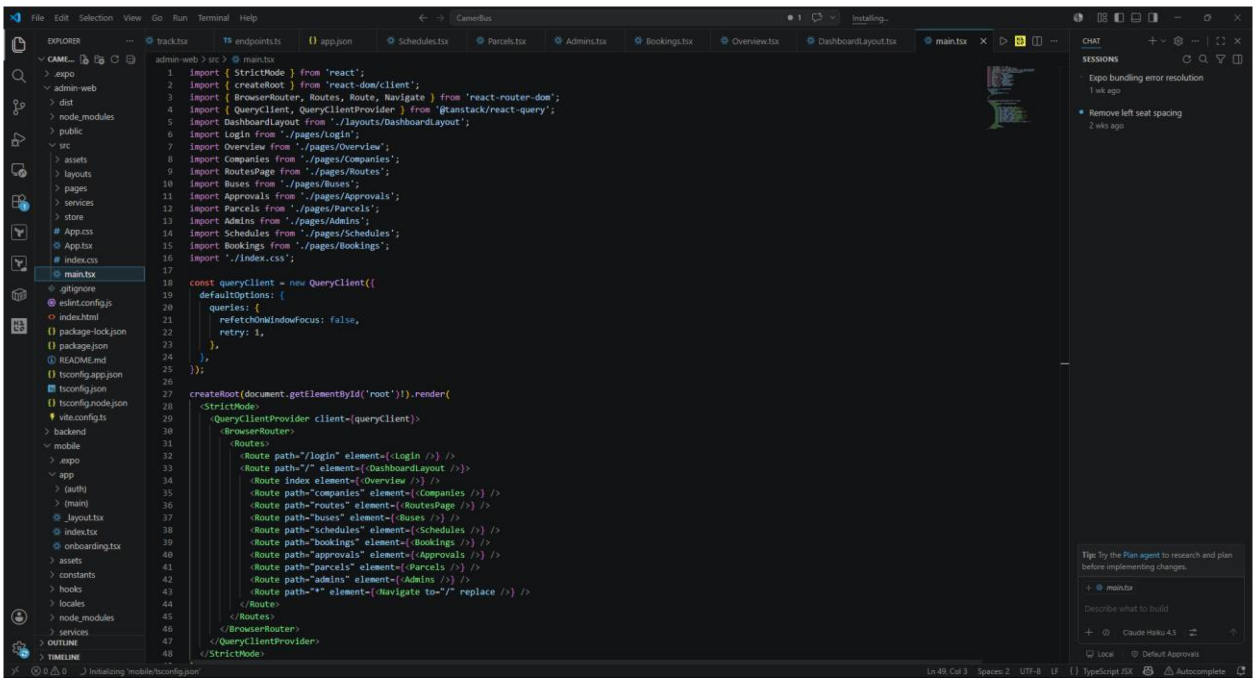


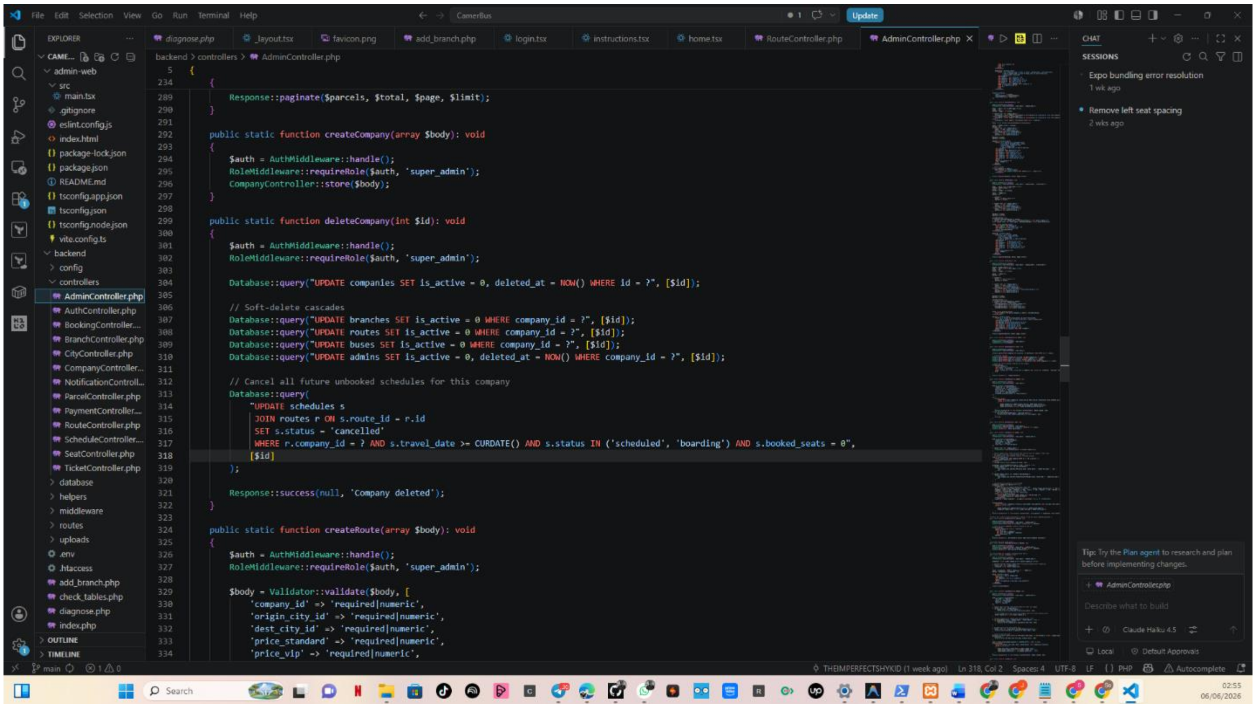
APPENDIX 2: Mobile app Extra pages

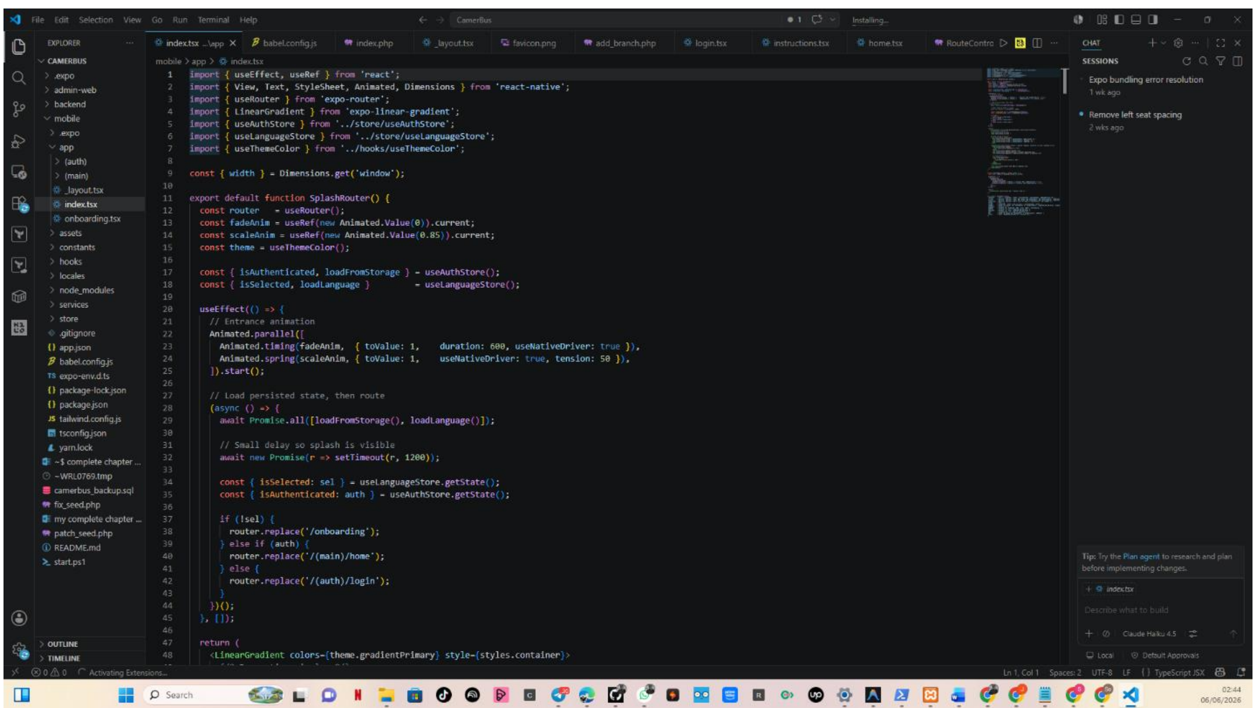




APPENDIX 3: Source code screenshots




backend > controllers > AdminController.php
Response::paginate($parcels, $total, $page, $limit);
}
public static function createCompany(array $body): void
{
$auth = AuthMiddleware::handle();
RoleMiddleware::requireRole($auth, 'super_admin');
CompanyController::store($body);
}
public static function deleteCompany(int $id): void
{
$auth = AuthMiddleware::handle();
RoleMiddleware::requireRole($auth, 'super_admin');
Database::query("UPDATE companies SET is_active = 0, deleted_at = NOW() WHERE id = ?", [$id]);
// Soft-delete cascades
Database::query("UPDATE branches SET is_active = 0 WHERE company_id = ?", [$id]);
Database::query("UPDATE routes SET is_active = 0 WHERE company_id = ?", [$id]);
Database::query("UPDATE buses SET is_active = 0 WHERE company_id = ?", [$id]);
Database::query("UPDATE admins SET is_active = 0, deleted_at = NOW() WHERE company_id = ?", [$id]);
// Cancel all future unbooked schedules for this company
Database::query(
"UPDATE schedules s
JOIN routes r ON s.route_id = r.id
SET s.status = 'cancelled'
WHERE r.company_id = ? AND s.travel_date >= CURDATE() AND s.status IN ('scheduled', 'boarding') AND s.booked_seats = 0",
[$id]
);
Response::success(null, 'Company deleted');
}
public static function createRoute(array $body): void
{
$auth = AuthMiddleware::handle();
RoleMiddleware::requireRole($auth, 'super_admin');
$body = Validator::validate($body, [
'company_id' => 'required|numeric',
'origin_city_id' => 'required|numeric',
'dest_city_id' => 'required|numeric',
'price_standard' => 'required|numeric',
'price_vip' => 'required|numeric',


mobile > app > index.tsx
import { useEffect, useRef } from 'react';
import { View, Text, StyleSheet, Animated, Dimensions } from 'react-native';
import { useRouter } from 'expo-router';
import { LinearGradient } from 'expo-linear-gradient';
import { useAuthStore } from '../store/useAuthStore';
import { useLanguageStore } from '../store/useLanguageStore';
import { useThemeColor } from '../hooks/useThemeColor';
const { width } = Dimensions.get('window');
export default function SplashRouter() {
const router = useRouter();
const fadeAnim = useRef(new Animated.Value(0)).current;
const scaleAnim = useRef(new Animated.Value(0.85)).current;
const theme = useThemeColor();
const { isAuthenticated, loadFromStorage } = useAuthStore();
const { isSelected, loadLanguage } = useLanguageStore();
useEffect(() => {
// Entrance animation
Animated.parallel([
Animated.timing(fadeAnim, { toValue: 1, duration: 600, useNativeDriver: true }),
Animated.spring(scaleAnim, { toValue: 1, useNativeDriver: true, tension: 50 }),
]).start();
// Load persisted state, then route
(async () => {
await Promise.all([loadFromStorage(), loadLanguage()]);
// Small delay so splash is visible
await new Promise(r => setTimeout(r, 1200));
const { isSelected: sel } = useLanguageStore.getState();
const { isAuthenticated: auth } = useAuthStore.getState();
if (!sel) {
router.replace('/onboarding');
} else if (auth) {
router.replace('/(main)/home');
} else {
router.replace('/(auth)/login');
}
})();
}, []);
return (
<LinearGradient colors={theme.gradientPrimary} style={styles.container}>